



**DCL- QPSK**

**QPSK / DQPSK**

**MODULATION & DEMODULATION KIT**

**EXPERIMENTAL MANUAL**



**....A MESSAGE FROM**

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Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

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This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

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This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

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For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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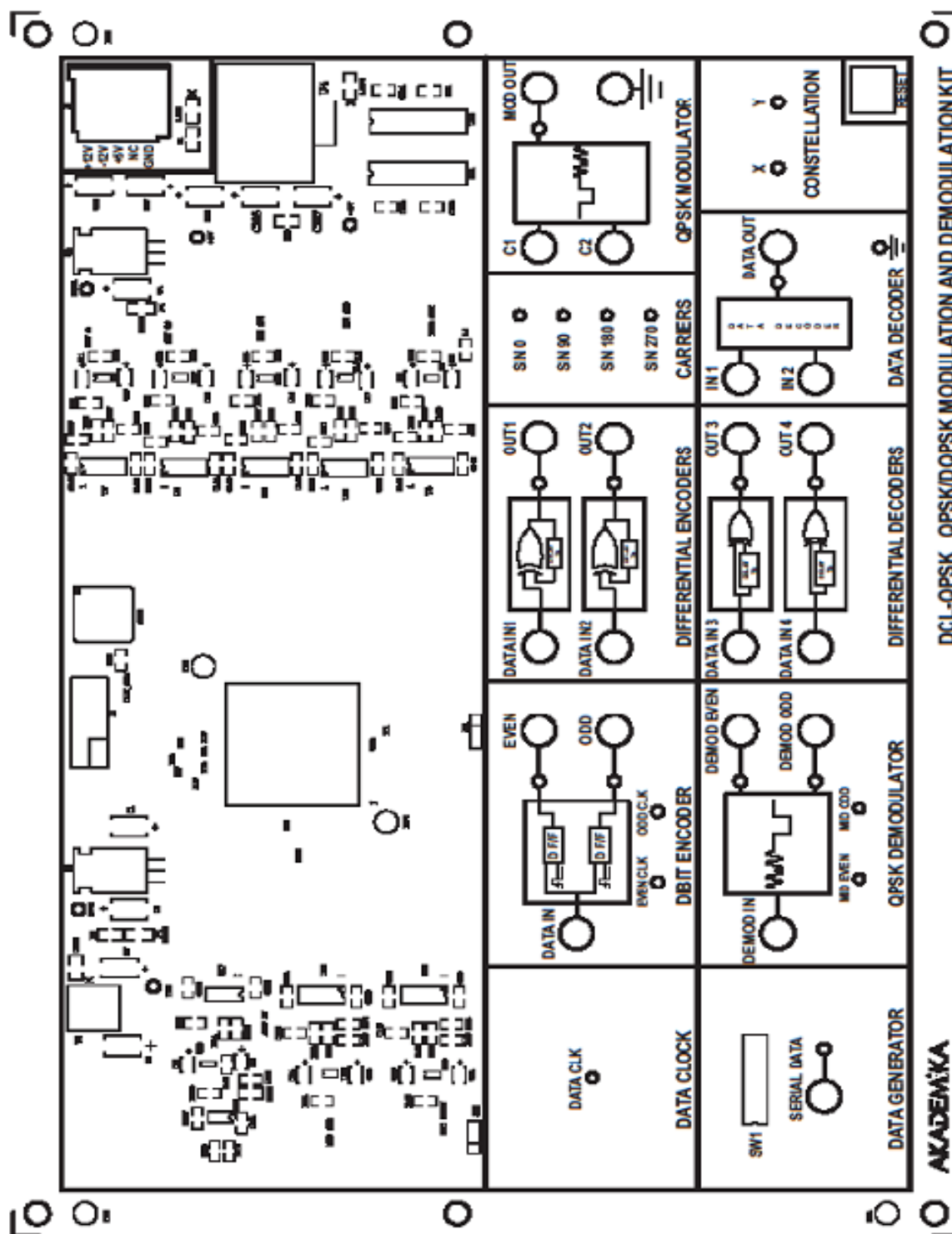
## TECHNICAL SPECIFICATIONS

### DCL-QPSK/DQPSK MODULATION AND DEMODULATION

|                                    |   |                                                                                                         |
|------------------------------------|---|---------------------------------------------------------------------------------------------------------|
| <b>DATA SIMULATOR</b>              | : | Onboard 8-bit variable NRZ-L pattern                                                                    |
| <b>CRYSTAL OSCILLATOR</b>          | : | 32.768 MHz                                                                                              |
| <b>CLOCK FREQUENCY</b>             | : | 256 KHz.                                                                                                |
| <b>DATA ENCODING</b>               | : | NRZ (L), Dibit data (Even data & Odd data), Differentially encoded Even data & Odd data.                |
| <b>ONBOARD CARRIER SINE WAVES:</b> |   | 1MHz (0°), 1MHz (90°), 1MHz (180°), 1MHz (270°)                                                         |
| <b>CARRIER MODULATION</b>          | : | QPSK, DQPSK                                                                                             |
| <b>CARRIER DEMODULATION</b>        | : | QPSK, DQPSK                                                                                             |
| <b>DATA DECODING</b>               | : | NRZ (L), Dibit data (Even data & Odd data), Differentially encoded Even data & Odd data.                |
| <b>CONSTELLATION DIAGRAM</b>       | : | Provision for observing Constellation diagram and the concept can be well understood.                   |
| <b>INTERMEDIATE SIGNAL</b>         | : | Provision for observing 2 intermediate signals during demodulation one for even and other for odd data. |
| <b>INTER CONNECTION</b>            | : | 2 mm banana socket.                                                                                     |
| <b>POWER SUPPLY</b>                | : | +12V, -12V, +5V, GND.                                                                                   |
| <b>TEST POINTS</b>                 | : | 28                                                                                                      |
| <b>ACCESSORIES</b>                 |   |                                                                                                         |

| DCL-QPSK |                                  |              | Quantity |
|----------|----------------------------------|--------------|----------|
| 1        | Short Links PCS2 Patch cord      | Male To Male | 10       |
| 2        | ADCL-02 EXPT Manual              |              | 1        |
| 3        | 5 PIN DIN CONNECTOR Power Supply |              | 1        |
| 4        | Power cord                       |              | 1        |

## FUNCTIONAL BLOCK



## **INTRODUCTION**

AKADEMIKA with the trainer, namely, “DCL-QPSK” initiates the user to QPSK/DQPSK MODULATION AND DEMODULATION technique adopted in practice

DCL-QPSK/DQPSK MODULATION AND DEMODULATION KIT

### **FEATURES:**

- 1) The trainer comes with an onboard data simulator, which generates the NRZ-L pattern depending on positions of 8-bit switch and data clock of 256 KHz.
- 2) VLSI based design.
- 3) Separate component and operational area.
- 4) Modular organization of circuit functions.
- 5) Test points to access signals at every stage of circuit operation.
- 6) Interconnection facilities:  
Sockets and patch chords are provided for on board connections.
- 7) Test Points:  
All relevant test points are brought-out for observations. Observations are carried out on DSO.
- 8) Power Supply:  
+12V, -12V, +5V, GND.

## PRINCIPLES OF ADVANCED DIGITAL MODULATION AND DEMODULATION TECHNIQUES

The Digital data in the systems can be encoded in several formats. All these waveforms can be broadly classified into the following four groups:

1. Non-return to Zero formats,
2. Return to Zero formats,
3. Phase encoded formats,
4. Multilevel binary formats.

It is common that some puzzling questions may arise to a beginner as to why there should be so many PCM Waveforms? And are there really so many unique applications necessitating such a variety of waveforms to represent the “ones” and “zeros”? The reason for the large selection relates to the difference in performances that characterizes each waveform. In choosing a coding scheme for a particular application, the common parameters worth examination are as follows:

### 1) DC Component:

The DC component associated with the data formats has to be minimized, if the system is to be AC coupled. The proper selection of the data format ensures optimum DC component that is associated with the signal power spectrum.

### 2) Self Clocking:

The presence of sufficient number of transition in the transmitted data enables the receiver to derive the clock from the data. Thus it is preferable to select a data format, which enables sufficient number of transitions in the transmitted data stream. For example, the biphase signals allow sufficient number of transitions in the transmitted data, compared to the corresponding Non-return to zero signals.

### 3) Band Width:

The proper selection of the coding enables to optimize the bandwidth requirement.

### 4) Noise Immunity:

The various PCM waveform types can be further characterized by the probability of bit error versus signal to noise ratio. For Example, the NRZ waveforms have better immunity to noise than the corresponding unipolar return to zero signal, which enables an error free transmission.

## FUNDAMENTALS OF CODING AND CARRIER MODULATION TECHNIQUES:

Digital Communication Systems represent the information by the binary digits, ‘Ones’ and ‘Zeros’. The basic principles of digital communication are Sampling, Multiplexing, Encoding, Error Control Coding, Data Conditioning, and Carrier Modulation etc. In coding we represent sample data pattern in different formats. In carrier modulation, carrier is modulated as per modulating signal.

## CARRIER MODULATION SCHEMES:

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., ‘ones’ and ‘zeros’. These digits are represented by discrete

voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation Techniques normally adopted in practice fall under three broad categories.

- a) Amplitude Shift Keying (ASK),
- b) Frequency Shift Keying (FSK),
- c) Phase Shift Keying (PSK).

#### a) AMPLITUDE SHIFT KEYING (ASK):

For all types of Carrier Modulation, the Carrier frequency should be at least 2 times base band of the modulating signal.

Assuming that the modulated carrier is a sine wave represented by the following equation.

$$C(t) = A(t) \cdot \cos(\omega t)$$

Where  $C(t)$  = Carrier sine wave.

$A(t)$  = Time varying amplitude

$\omega t$  = Time varying angle.

Now in amplitude shift keying, the carrier is being transmitted only when the modulating data is 'one' and when the data is 'zero' the carrier is rejected from transmission. Thus the resulting modulated output of this type of modulation can be represented as follows:

$$M(t) = r(t) \cdot C(t)$$

Where  $M(t)$  = Modulated Carrier

$r(t)$  = Time varying modulating data which is either 'one' or 'zero'

Now,

$$M(t) = A(t) \cdot \cos(\omega t) \quad \text{when modulating data is one.}$$

$$= 0 \quad \text{when modulating data is zero.}$$

An Envelope detector is used to recover the data from the modulated carrier.

#### b. FREQUENCY SHIFT KEYING (FSK):

In this type of modulation, the modulated output shifts between two frequencies for all 'one' to 'zero' transitions.

Let the carrier frequencies be represented by  $\omega_1$  and  $\omega_2$ , and then we have:

$$M(t) = A(t) \cdot \cos(\omega_1 t) \quad \text{if data is 'One'}$$

$$= A(t) \cdot \cos(\omega_2 t) \quad \text{if data is 'Zero'}$$

Where  $A(t)$  = Time varying amplitude of the sine wave.

$M(t)$  = Modulated carrier.

FSK Demodulator employs PLL logic for the recovery of data.

#### c. PHASE SHIFT KEYING (PSK):

In the PSK modulation or phase shift keying, for all 'one' to 'zero' transitions of the modulating data, the modulated output switches between the in phase and out of phase components of the modulating frequency. If the modulated carrier is represented by:

$$\begin{aligned}
 M(t) &= A(t) \cdot \cos(wt + \text{Phase}) \\
 \text{Where } A(t) &= \text{Time varying amplitude, and} \\
 wt &= \text{Time varying angle.} \\
 M(t) &= \text{Modulated carrier.}
 \end{aligned}$$

Then the phase equals  $0^\circ$  whenever the data equals 'one' and the phase equals  $180^\circ$  whenever the data equals 'zero'. This type of phase shift keying is called Binary phase Shift Keying (BPSK).

There are also other forms of phase shift keying like Differential Phase Shift Keying (DPSK), Differentially Encoded Phase Shift Keying (DEPSK) Quadrature Phase Shift Keying (QPSK), and Differential Quadrature Phase Shift Keying (DQPSK) etc.

In DPSK/DEPSK the data stream to be transmitted is encoded in such a way to get the differentially encoded data as input to DPSK/DEPSK modulator. Then the phase equals  $0^\circ$  whenever the data equals 'one' and the phase equals  $180^\circ$  whenever the data equals 'zero'. This type of phase shift keying is called Differential phase Shift Keying (DPSK)/ differentially encoded phase Shift Keying (DEPSK).

In QPSK, phase has 4 different values like  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$  and  $270^\circ$ . They are used in the carrier modulation of differentially coded dibit pair. One of the four carriers is transmitted at a time with respect to the symbol generated by EVEN&ODD.

In DQPSK the inputs to the modulator (EVEN&ODD) are differentially encoded before applying to the modulator & then transmitted similar to the QPSK. At the receiver, the data is decoded before applying to the data decoder.

The BPSK/DPSK/DEPSK detector works on the principle of square law. BPSK/DPSK/DEPSK modulated carrier can be represented mathematically as:

$$\begin{aligned}
 M(t) &= A(t) \cdot \cos(wt + \text{Phase}) \\
 \text{We have phase} &= 0^\circ && \text{- when data is 1} \\
 &= 180^\circ && \text{- when data is 0} \\
 \text{Thus,} \\
 M(t) &= A(t) \cdot \cos(wt) && \text{- when data is 1} \\
 &= -A(t) \cdot \cos(wt) && \text{- when data is 0}
 \end{aligned}$$

On squaring the modulated carrier, the negative sign gets eliminated and the frequency gets multiplied by two.

$$\begin{aligned}
 M^2(t) &= A^2(t) \cdot \cos^2(wt) && \text{- for both data is 1 and 0} \\
 M^2(t) &= A^2(t) \cdot (1 + \cos(2wt))/2
 \end{aligned}$$

Now the in phase reference carrier can be recovered by dividing the frequency of the squared modulated carrier by two.

Once the carrier is recovered, the data can be detected by comparing the phase of the received modulated carrier with the phase of the reference carrier. In Quadrature Phase

Shift Keying (QPSK), as with binary PSK, information carried by the transmitted signal is contained in the phase. In particular, the phase of the carrier takes on one of four equally spaced values, such as 0, 90, 180 and 270 degree for this set of values we may define the transmitted signal as

$$S_i(t) = \begin{cases} \sqrt{\frac{2E}{T}} \cos \left[ 2\pi f_c t + (2i-1) \frac{\pi}{4} \right], & 0 \leq t \leq T \\ 0, & \text{elsewhere} \end{cases} \quad \text{EQ. 1}$$

Where  $i = 1, 2, 3, 4$ ;  $E$  is transmitted signal energy per symbol, and  $T$  is the symbol duration. The carrier frequency  $f_c$  equals  $n_c/T$  for some fixed integer  $n_c$ . Each possible value of the phase corresponds to a unique dibit. Thus, for example, we may choose the foregoing sets of values to represent the gray encoded set of dibits: 10, 00, 01, 11 where only a single bit is changed from one dibit to the next.

Using a well-known trigonometric identity, we may use the equation 1 to redefine the transmitted signal  $S_i(t)$  for the interval  $0 \leq t \leq T$  in the equivalent form:

$$S_i(t) = \frac{2E}{T} \cos(2i-1) \frac{\pi}{4} \cos(2\pi f_c t) - \frac{2E}{T} \sin(2i-1) \frac{\pi}{4} \sin(2\pi f_c t)$$

Where  $i = 1, 2, 3, 4$ . based on this representation, we can make the following observation:

There are two orthonormal basis functions,  $\phi_1(t)$  and  $\phi_2(t)$ , contained in the expansion of  $S_i(t)$ . Specifically,  $\phi_1(t)$  and  $\phi_2(t)$  are defined by a pair of quadrature carriers:

$$\phi_1(t) = \frac{2}{T} \cos(2\pi f_c t), \quad 0 \leq t \leq T$$

$$\phi_2(t) = \frac{2}{T} \sin(2\pi f_c t), \quad 0 \leq t \leq T$$

There are four message points, and the associated signal vectors are defined by

$$S_i = \begin{bmatrix} \sqrt{E} \cos \left[ (2i-1) \frac{\pi}{4} \right] \\ -\sqrt{E} \sin \left[ (2i-1) \frac{\pi}{4} \right] \end{bmatrix}, \quad i = 1, 2, 3, 4$$

Accordingly, a QPSK signal has a two-dimensional signal constellation (i.e.  $M=4$ ) and four message points (i.e.  $M=4$ ) whose phase angles increase in a counter clockwise direction. As with binary PSK, the QPSK signal has minimum average energy.

## Circuit Description

- **Clock And Data Generator**

This block generates NRZ-L data of a variable pattern using VLSI technology depending on positions of eight bit switch SW1 also Reference clock of frequency 256KHz (DATA CLK) is generated. The crystal oscillator generates a 32.768MHz clock, from which a 256 KHz data clock is generated by using downcounter. For data generation output of SW1 is given to Parallel to serial shift register. Output of shift register is taken as Reference Data.

- **Dibit Encoder Section**

The logic for the dibit encoder is written in VHDL and the serial data is split into even data and odd data based on the clock of 128 KHz known as even clock and odd clock.

- **Differential Encoder Section**

The even and odd data is differentially encoded using a differential encoder the logic for which is written in VHDL and the differential encoded even and odd data can be observed on out1 out2 port.

- **Carriers Section**

All the carriers are generated inside the FPGA using VHDL which is then amplified, level shifted and given to DAC convertor. The carriers are of frequency 1MHz and with a phase difference of  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$  and  $270^\circ$ . The carriers can be observed at the respective test point in the carriers section.

- **QPSK Modulator Section**

The modulation is done in this section where for each symbol of two bits a carrier of specific phase is transmitted. The entire logic is written in VHDL.

- **QPSK Demodulator Section**

The demodulator section consists of a multiplier and filter. On receiving the modulated signal it is first multiplied so that the phase difference is removed and then filtered so that the actual even and odd data or when differential encoder is used in the modulation section then differential encoded even and odd data, can be recovered. The logic for this is written in VHDL.

- **Differential Decoder Section**

When differential encoder is used in the modulation section then differential decoder is used in demodulator section so that actual even and odd data can be recovered from differentially encoded even and odd data. The logic for this is written in VHDL.

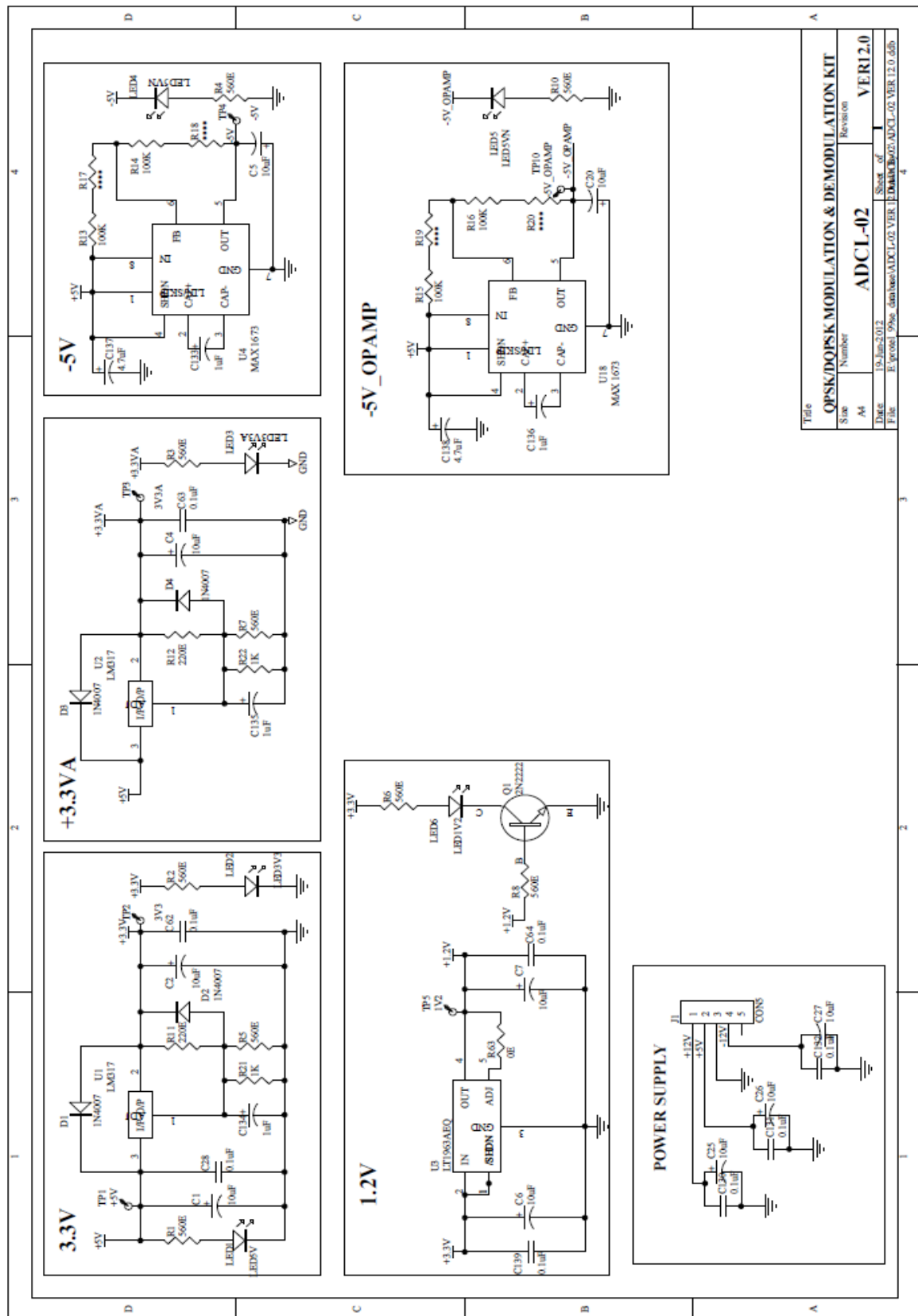
- **Data Decoder Section**

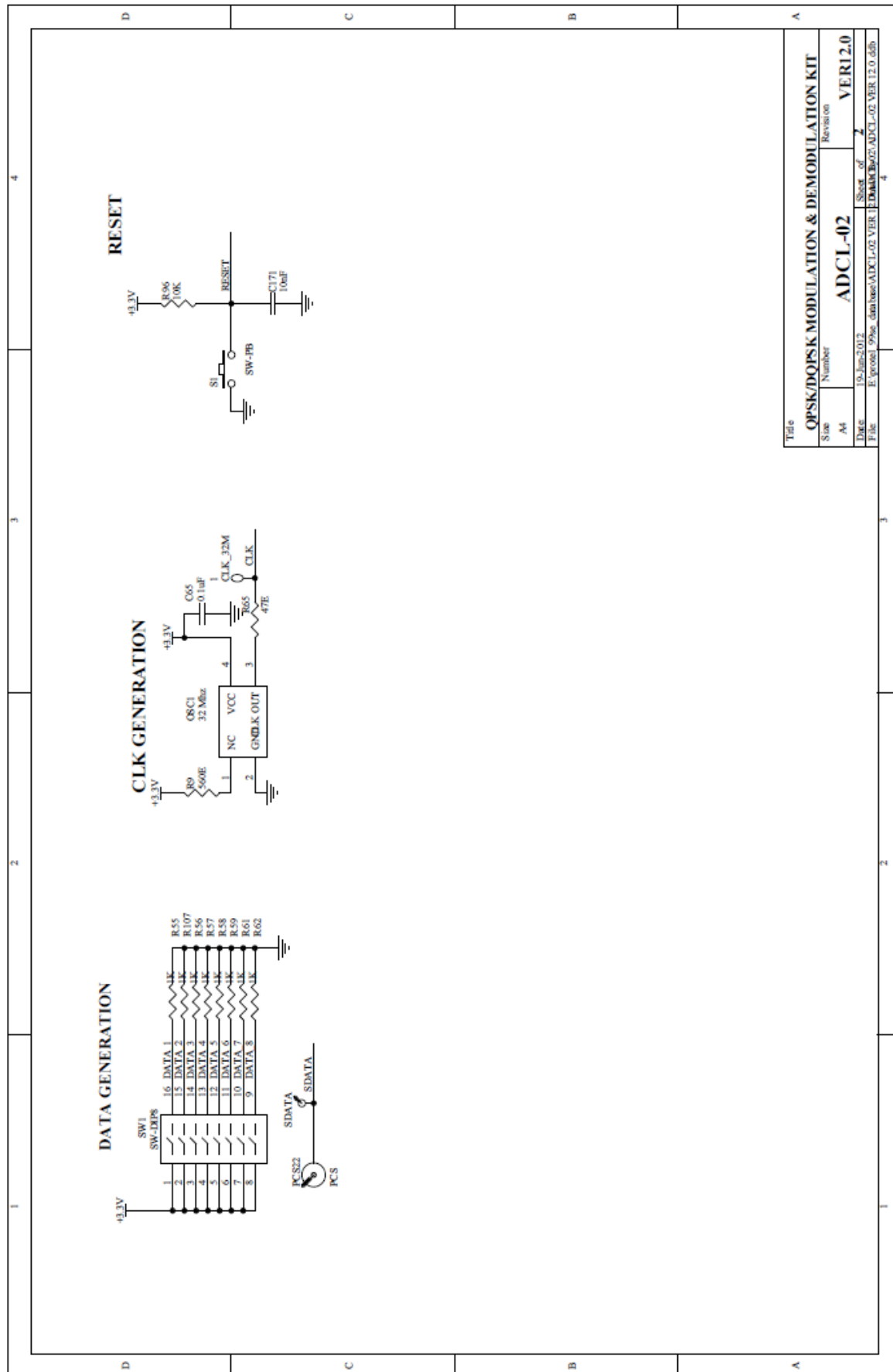
This section consists of a 2:1 Mux where we get the input serial data from even and odd data. The logic is written in VHDL.

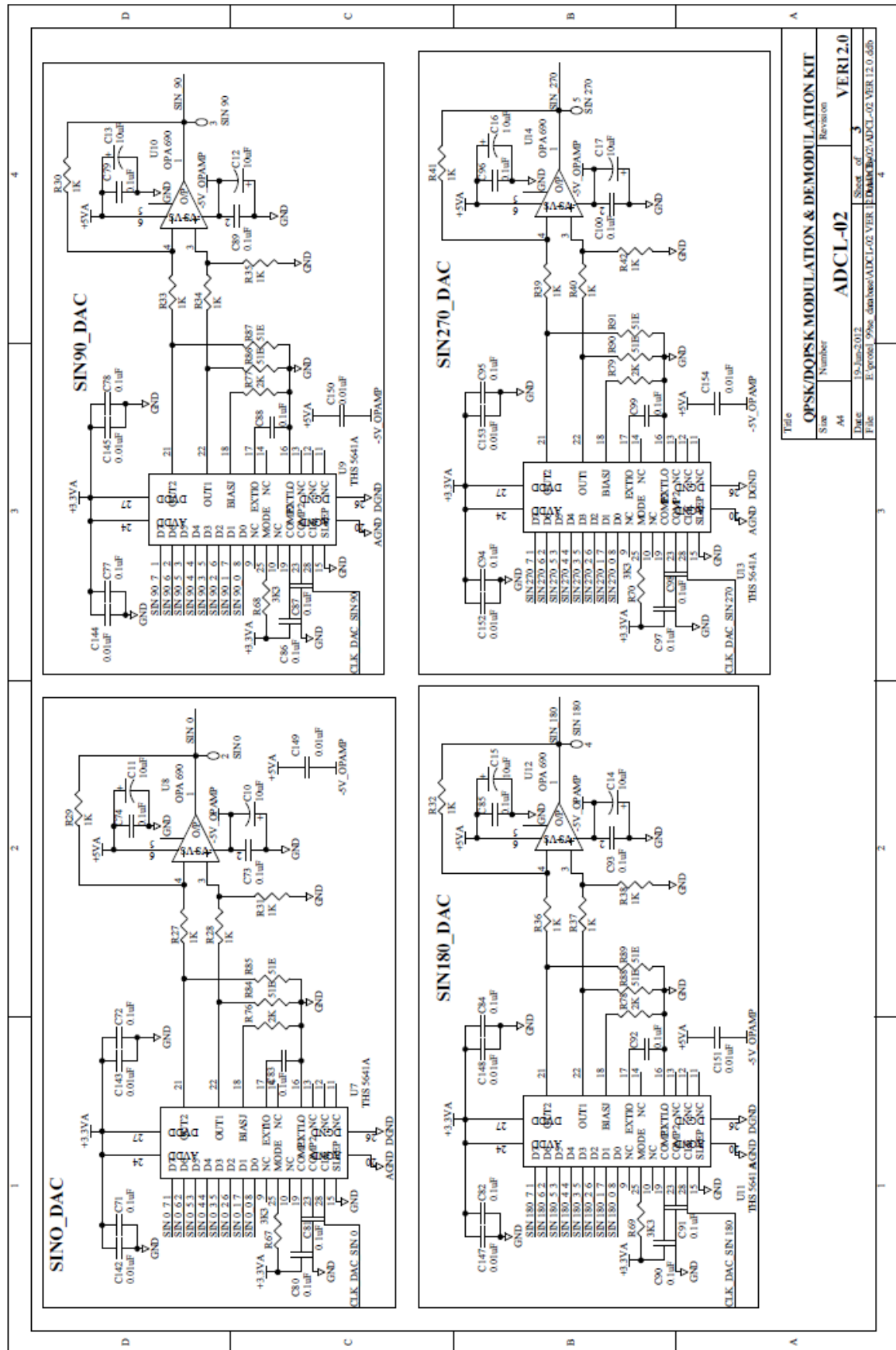
- **Constellation section:**

The constellation diagram of even and odd data can be observed here. Two multiplexer/de-multiplexer IC 4051 is used for this purpose.

## CIRCUIT DIAGRAM OF DCL-QPSK

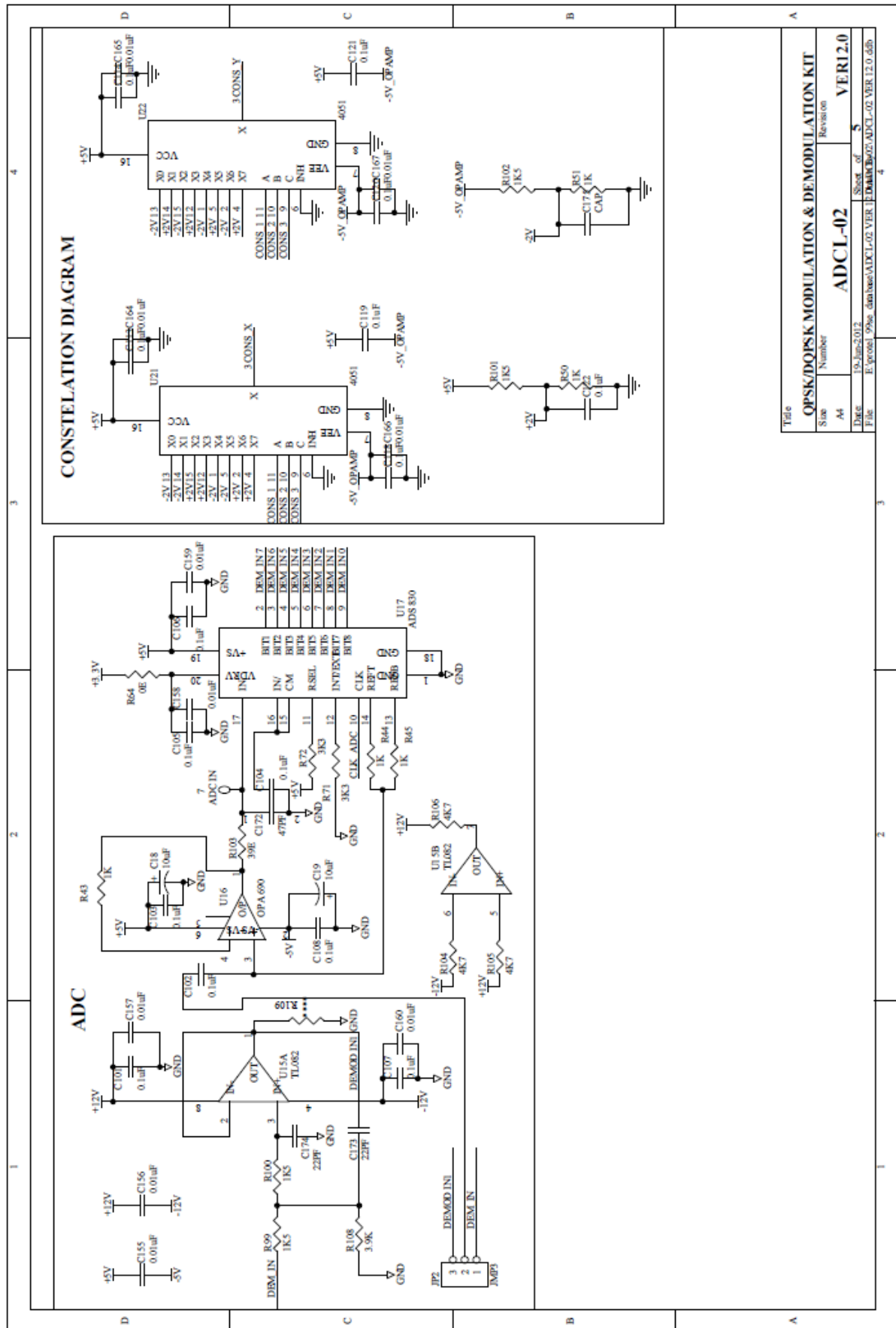




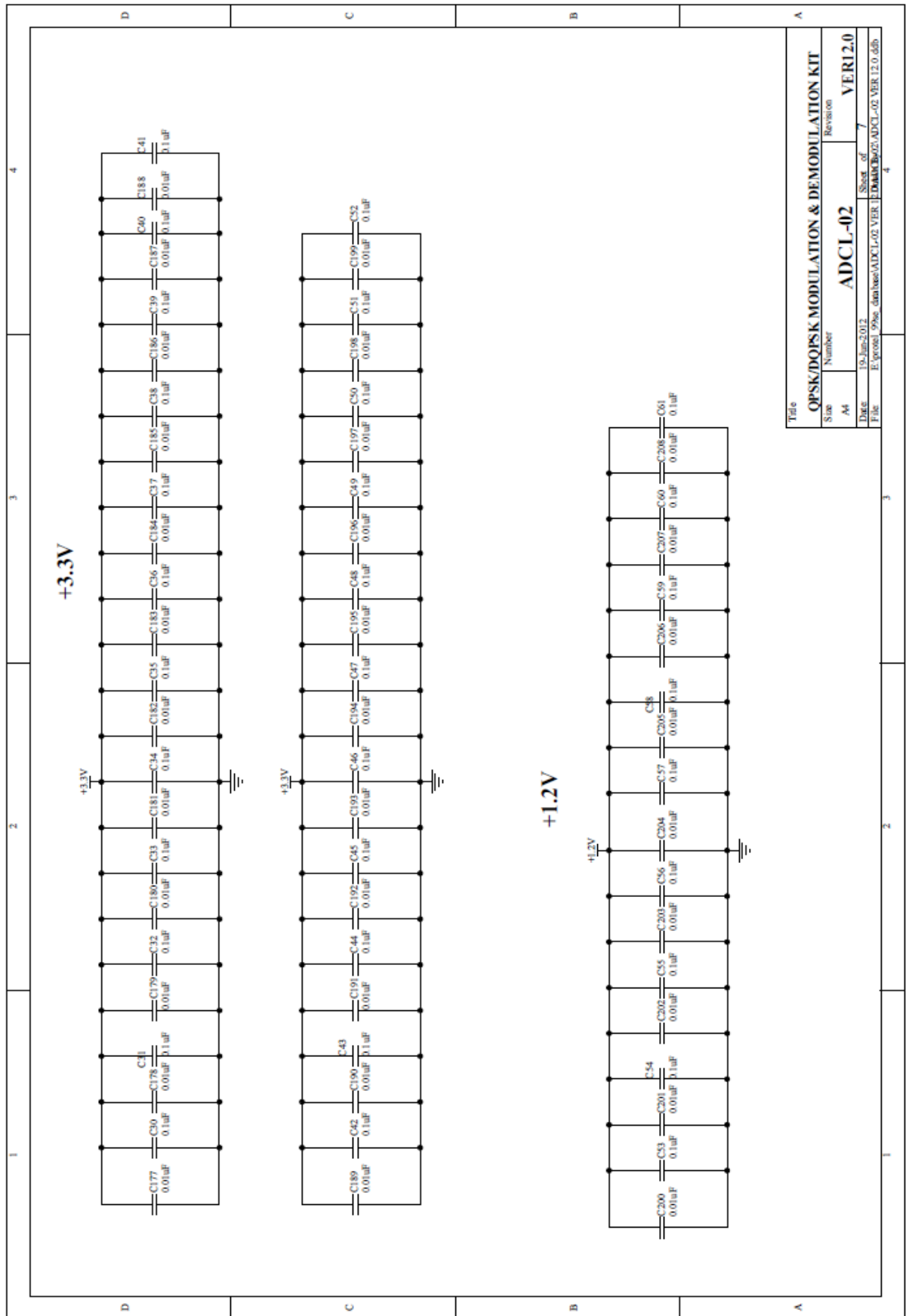


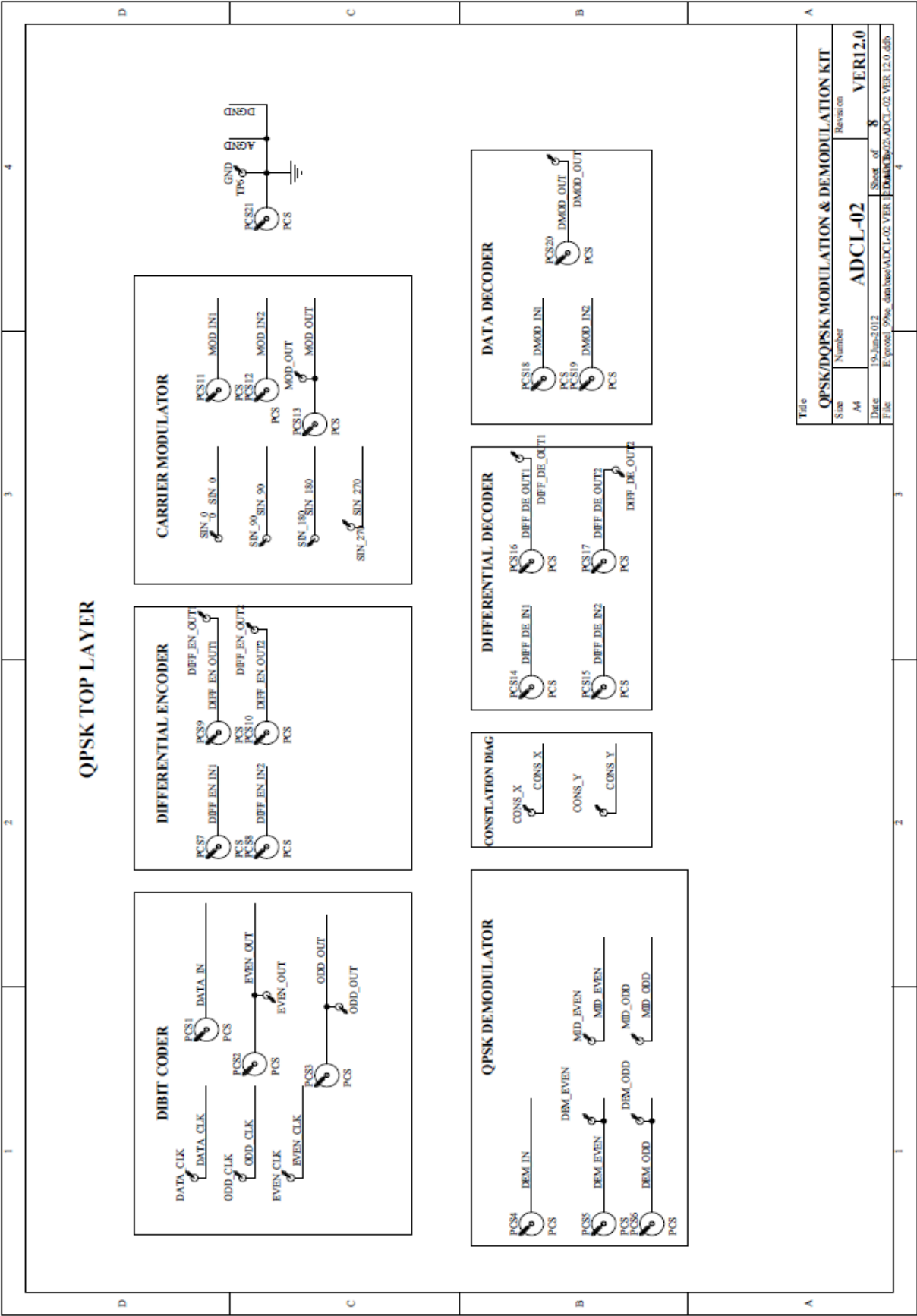
| Title                                    |                                                     |          |   |
|------------------------------------------|-----------------------------------------------------|----------|---|
| QPSK/DQPSK MODULATION & DEMODULATION KIT |                                                     |          |   |
| State                                    | Number                                              | Revision |   |
|                                          | ADCL-02                                             | VER1.0   |   |
| Date                                     | 19-Jan-2012                                         | Sheet of | 3 |
| File                                     | E:\proj\99s\adcl\adcl-02\VER1.0\ADCL-02_VBR12.0.dbs |          |   |











# **EXPERIMENT NO. 1**

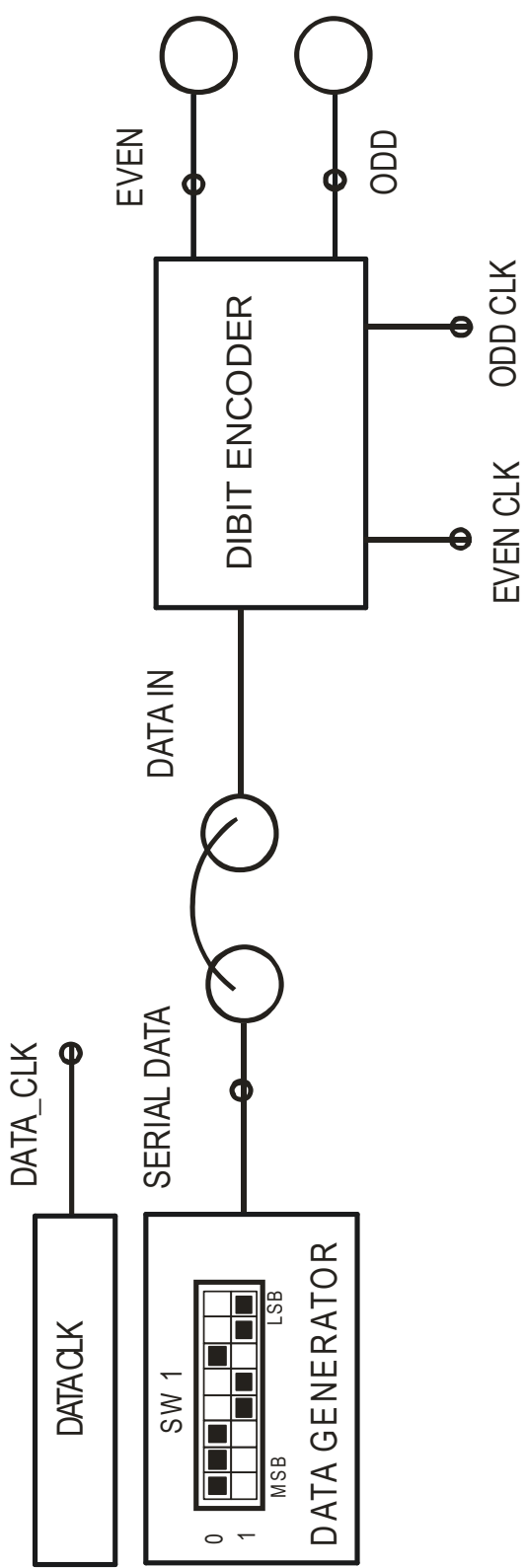


FIG 1.1 BLOCK DIAGRAM FOR DIBIT PAIR ENCODING OF NRZ-L DATA

## **EXPERIMENT NO: 1**

### **NAME:**

DIBIT PAIR DATA CODING TECHNIQUE OF NRZ-L DATA FORMAT

### **OBJECTIVE:**

To study dibit pair coding techniques for Non-Return to Zero data format

### **THEORY:**

NON - RETURN TO ZERO signal are the easiest formats that can be generated. These signals do not return to zero with the clock. The frequency component associated with these signals are half that of the clock frequency. Non-return to zero encoding is commonly used in slow speed communications interfaces for both synchronous and asynchronous transmission. Using NRZ, logic 1 bit is sent as a high value and logic 0 bit is sent as a low value.

The mechanism by which a bit stream  $b(t)$  generates a QPSK signal for transmission is shown in fig. 1.2 And relevant waveforms are shown in fig. 1.3. In these waveforms we have arbitrarily assumed that in every case the active edge is the upward edge. The toggle flip-flop generates an odd clock waveform and an even waveform. These clocks have period  $2T_b$ . The active edge of one of the clocks and the active edge of the other are separated by the bit time  $T_b$ . The bit stream  $b(t)$  is applied as the data input to type-D flip-flops, one driven by the odd and one driven by the even clock waveform. The flip-flop register alternates bits in the stream  $b(t)$  and hold each such registered bit for two bit intervals that is for the time  $2T_b$ . In fig. 1.4. We have numbered the bits in the stream  $b(t)$ . Note that the bit stream  $B_o(t)$  (which is the output of the flip-flop driven by the odd clock) registers bit 1 and holds that bit for time  $2T_b$ , then register bit 3 for time  $2T_b$ , then bit 5 for  $2T_b$ , etc. The even bit stream  $B_e(t)$  holds, for times  $2T_b$  each, the alternates bits numbered 2, 4, 6, etc.

### **EQUIPMENTS:**

- DCL-QPSK/DQPSK Kit
- Connecting Chords
- Power supply
- DSO

### **PROCEDURE:**

1. Refer to the block diagram (Fig. 1.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **DIBIT ENCODER**.
5. Observe **EVEN CLK**, **ODD CLK** and the coded signal **EVEN data** & **ODD data** on the oscilloscope with respect to **SERIAL DATA**.

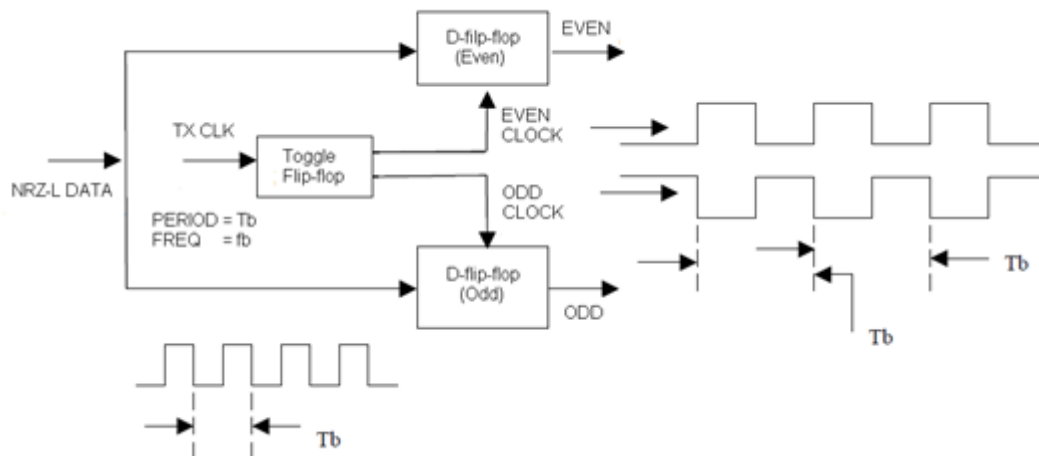


FIG. 1.2 DIBIT CONVERSION

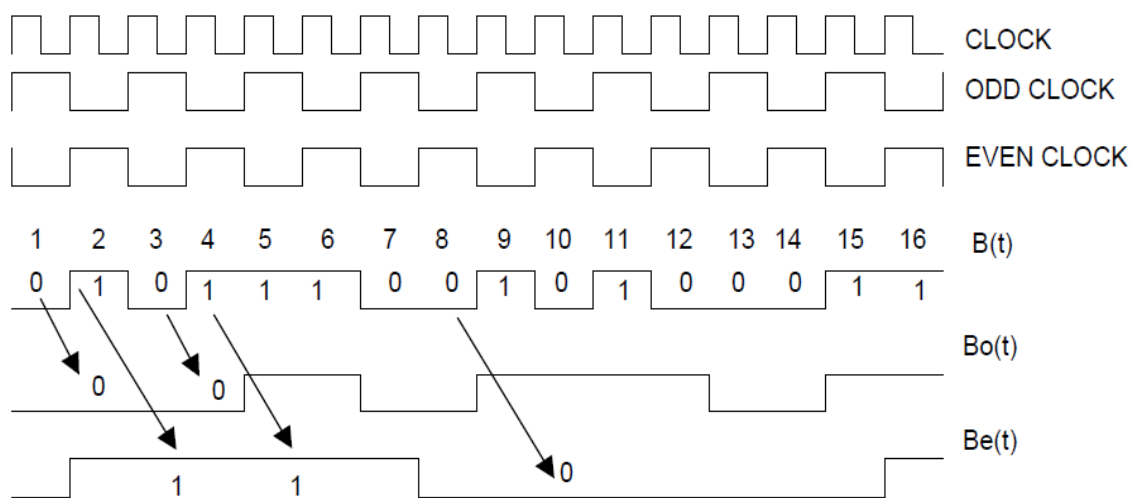


FIG. 1.3 WAVEFORM FOR DIBIT CONVERSION

**OBSERVATION:**

Observe the following waveforms on oscilloscope and plot it on the paper.

1. Transmitter clock **DATA\_CLOCK** (Fig. 1.4 (a))
2. NRZ-L coded data **SERIAL DATA** (Fig. 1.4 (a))
3. **EVEN CLK** with respect to **ODD CLK** (Fig. 1.4 (b))
4. Coded data **EVEN** with respect to **EVEN CLK** (Fig. 1.4 (c))
6. Coded data **ODD** with respect to **ODD CLK** (Fig. 1.4 (d))

### CAPTURED WAVEFORM:

**CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)**

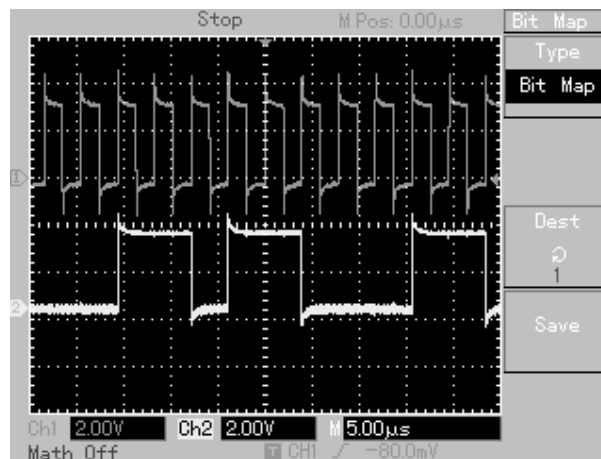


FIG. 1.4(a)

**CH 1: EVEN CLK (128 KHz) & CH 2: ODD CLK (128 KHz)**

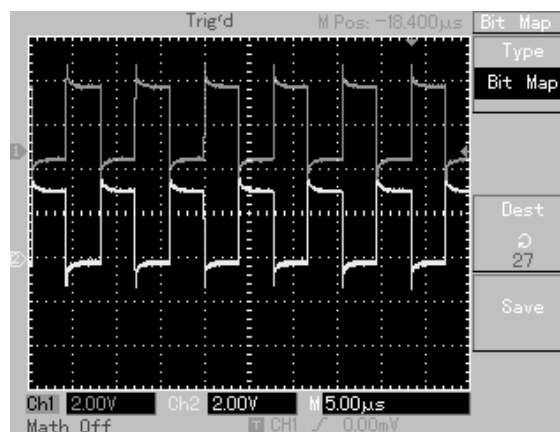
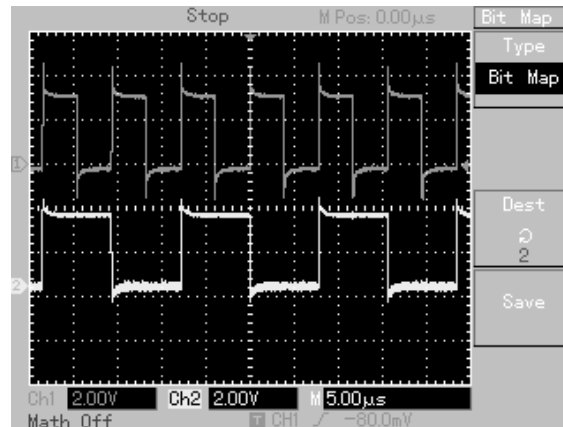
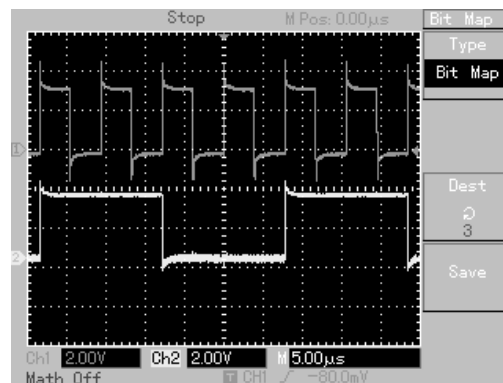


FIG. 1.4(b)

**CH 1: EVEN CLK (128 KHz) & CH 2: EVEN DATA****FIG. 1.4(c)****CH 1: ODD CLK (128 KHz) & CH 2: ODD DATA****FIG. 1.4(d)****CONCLUSION:**

The data are coded in dibit which generates,

- a) A data signal **EVEN** (in phase) consisting in voltage levels corresponding to the value of the first bit of the considered pair, for a period equal to 2 bit intervals.
- b) A data signal **ODD** (in quadrature) consisting in voltage levels corresponding to the value of the second bit of the pair, for duration equal to 2 bit intervals.

# **EXPERIMENT NO. 2**

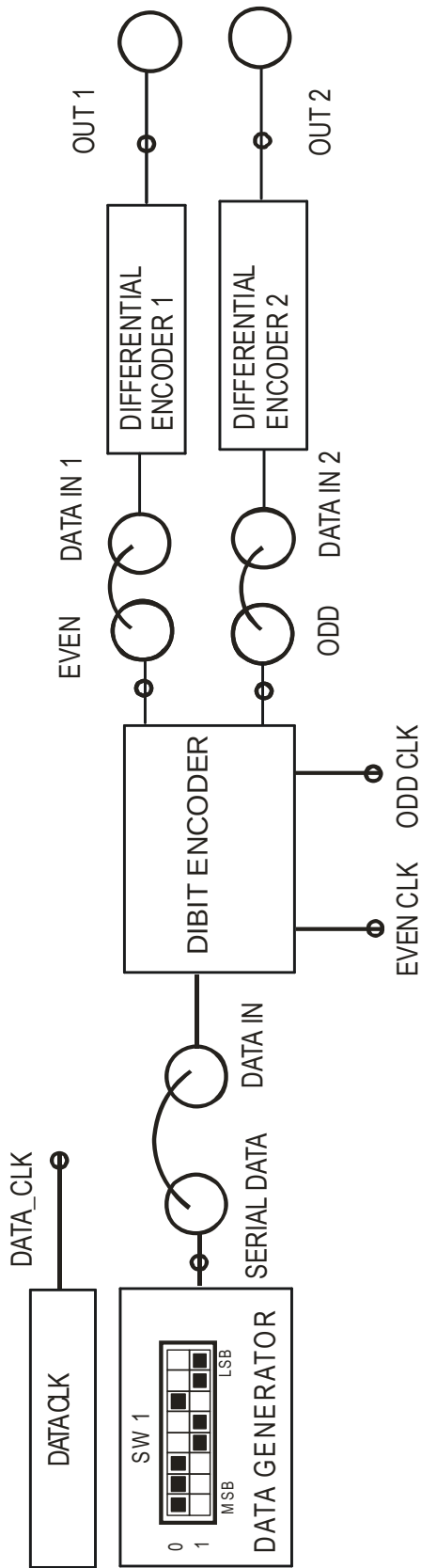


FIG 2.1 BLOCK DIAGRAM FOR  
STUDY OF DIFFERENTIAL ENCODING OF DIBIT PAIR DATA

## **EXPERIMENT NO: 2**

### **NAME:**

### **DIFFERENTIAL ENCODING OF DIBIT DATA**

### **OBJECTIVE:**

To study differential encoding techniques for dibit data

### **THEORY:**

#### **Differential encoding logic:**

A means of generating a differentially encoded data is shown in fig.2.2. The data stream,  $D(t)$ , to be transmitted is applied to one input of an exclusive-OR logic gate and to the other input ( $t-T_b$ ) is applied which is nothing but the output of the exclusive-OR gate  $B(t)$  delayed by the time  $T_b$  allocated to one bit.

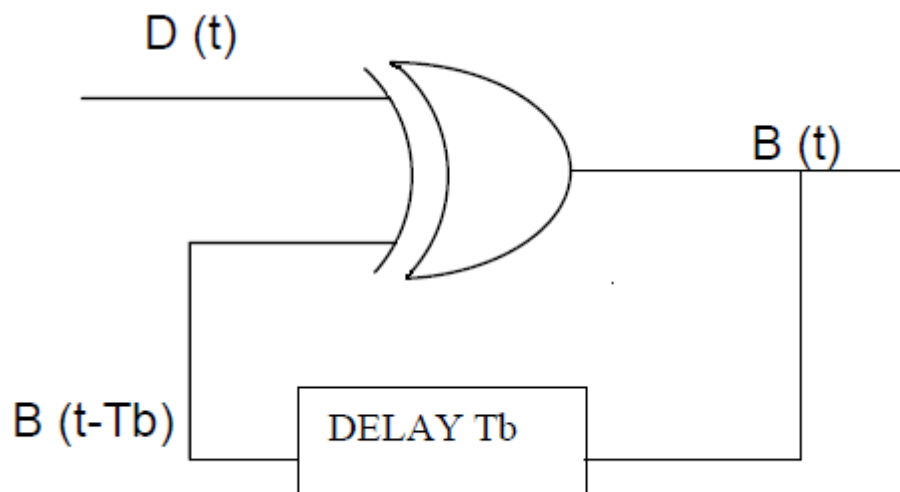


FIG. 2.2 DIFFERENTIAL ENCODER

In fig. 2.3 we have drawn the logic waveform to illustrate the response  $b(t)$  to an input  $d(t)$ . The upper level of the waveform corresponds to logic 1, the lower level to logic 0. Because of the feedback involved in the system of fig 2.3 there is difficulty in determining the logic levels in the interval in which we start to draw the Waveform (interval 1 in fig. 2.3) we cannot determine  $b(t)$  in this first interval of our waveform unless we know  $b(k=0)$ . But we cannot determine  $b(0)$  unless we know both  $d(0)$  and  $b(-1)$ , etc. thus, to justify any set of logic levels in an initial bit interval we need to know the logic levels in the preceding interval. But such a determination requires information about the interval two bit times earlier and so on. In the waveform of fig.2.3 we have circumvented the problem by arbitrarily assuming that in the first interval  $b(0)=0$ . It is shown below that, the data will be correctly determined regardless of our assumption concerning  $b(0)$ .

We now observe that the response of  $b(t)$  to  $d(t)$  is that  $b(t)$  changes level at the beginning of each interval in which  $d(t)=1$  and  $b(t)$  does not change level when  $d(t)=0$ . Thus during interval 3,  $d(3)=1$ , and correspondingly  $b(3)$  changes at the beginning of that interval.

During interval 6 & 7,  $d(6) = d(7) = 1$  and there are changes in  $b(t)$  at the beginning of both intervals. During bits 10, 11, 12 and 13  $d(t) = 1$  and there are changes in  $b(t)$  at the beginnings of each of these intervals. This behavior is to be anticipated from the truth tables of the exclusive-OR gate. For we note that when  $d(t) = 0$ ,  $b(t) = b(t-T_b)$  so that, whatever the initial value of  $b(t-T_b)$ , it reproduces itself. On the other hand when  $d(t) = 1$  then  $b(t) = 1/b(t-T_b)$ . Thus, in each successive bit intervals  $b(t)$  changes from its value in the previous interval.

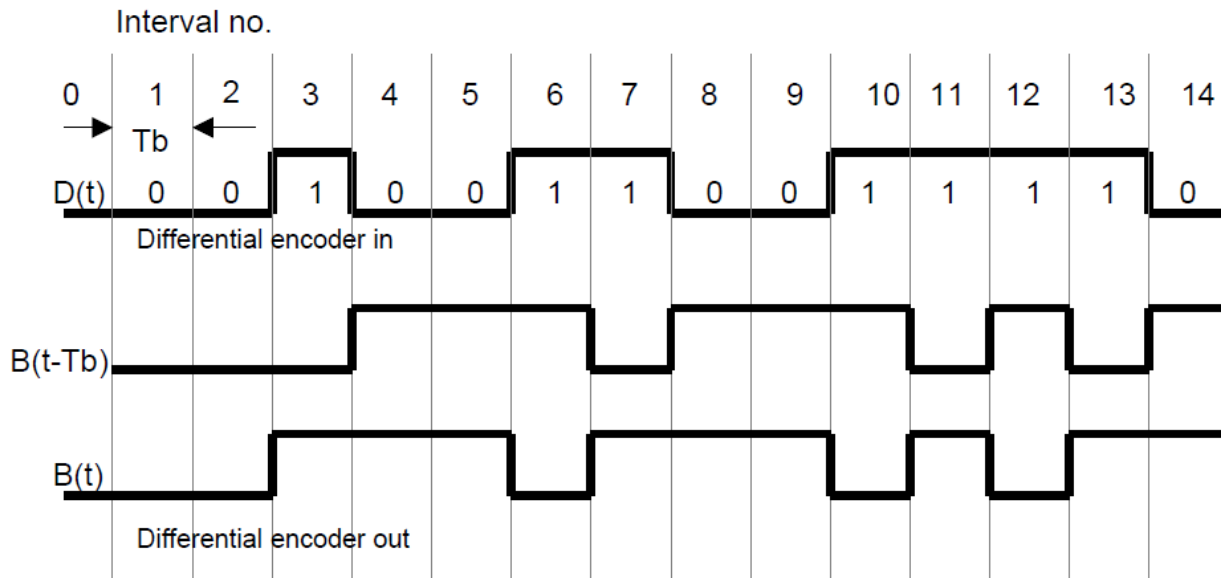


FIG. 2.3 WAVEFORM OF DIFFERENTIAL ENCODING OF DATA

Note that in some intervals where  $d(t) = 0$  we have  $b(t) = 0$  and in other intervals when  $d(t) = 0$  we have  $b(t) = 1$ . Similarly, when  $d(t) = 1$  sometimes  $b(t) = 1$  and sometimes  $b(t) = 0$ . Thus there is no correspondence between the levels of  $d(t)$  and  $b(t)$ , and the only invariant feature of the system is that a change (sometime up and sometime down) in  $b(t)$  occurs whenever  $d(t) = 1$ , and that no change in  $b(t)$  will occur whenever  $d(t) = 0$ .

Finally, we note that in waveform of fig. 2.3 are drawn on the assumption that, in interval 1,  $b(0) = 0$ . As is easily verified, if not intuitively apparent, if we had assumed  $b(0) = 1$ , the invariant feature by which we have characterized the system would continue to apply. Since  $b(0)$  must be either  $b(0) = 0$  or  $b(0) = 1$ , there being no other possibilities, our result is valid quite generally. If, however, we had started with  $b(0) = 1$  the levels  $b(1)$  and  $b(0)$  would have been inverted.

## EQUIPMENTS:

- DCL-QPSK Kit
- Connecting Chords
- Power supply
- DSO

## PROCEDURE:

1. Refer to the block diagram (Fig.2.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated to the **DATA IN** of the **DIBIT ENCODER**.
5. Connect **EVEN** data to **DATA IN 1** of Differential Encoder 1 and **ODD** data to **DATA IN 2** of Differential Encoder 2.
6. Observe the input data and the differentially encoded EVEN & ODD data at **OUT1** and **OUT2** of respective Differential Encoders.

## OBSERVATION:

ON KIT **DCL-QPSK** observe the following waveforms on oscilloscope and plot it on the paper.

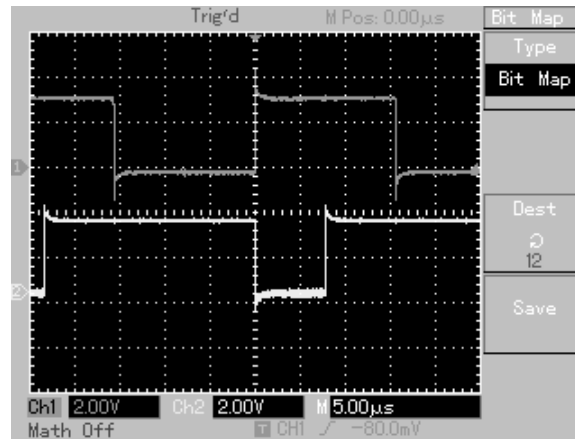
1. Differentially encoded data **OUT1** with respect to coded data **EVEN** (Fig.2.4(a))
2. Differentially encoded data **OUT2** with respect to coded data **ODD** (Fig.2.4(b))

## CAPTURED WAVEFORM:

**CH 1: EVEN & CH 2: OUT1**



**Fig.2.4 (a)**

**CH 1: ODD & CH 2: OUT2****Fig.2.4 (b)****CONCLUSION:**

It is observed that when input data to the differential encoder contains “1” the output of the differential encoder makes transition from its previous state. If input data contains “0” then output of differential encoder maintains the previous state.

# **EXPERIMENT NO. 3**

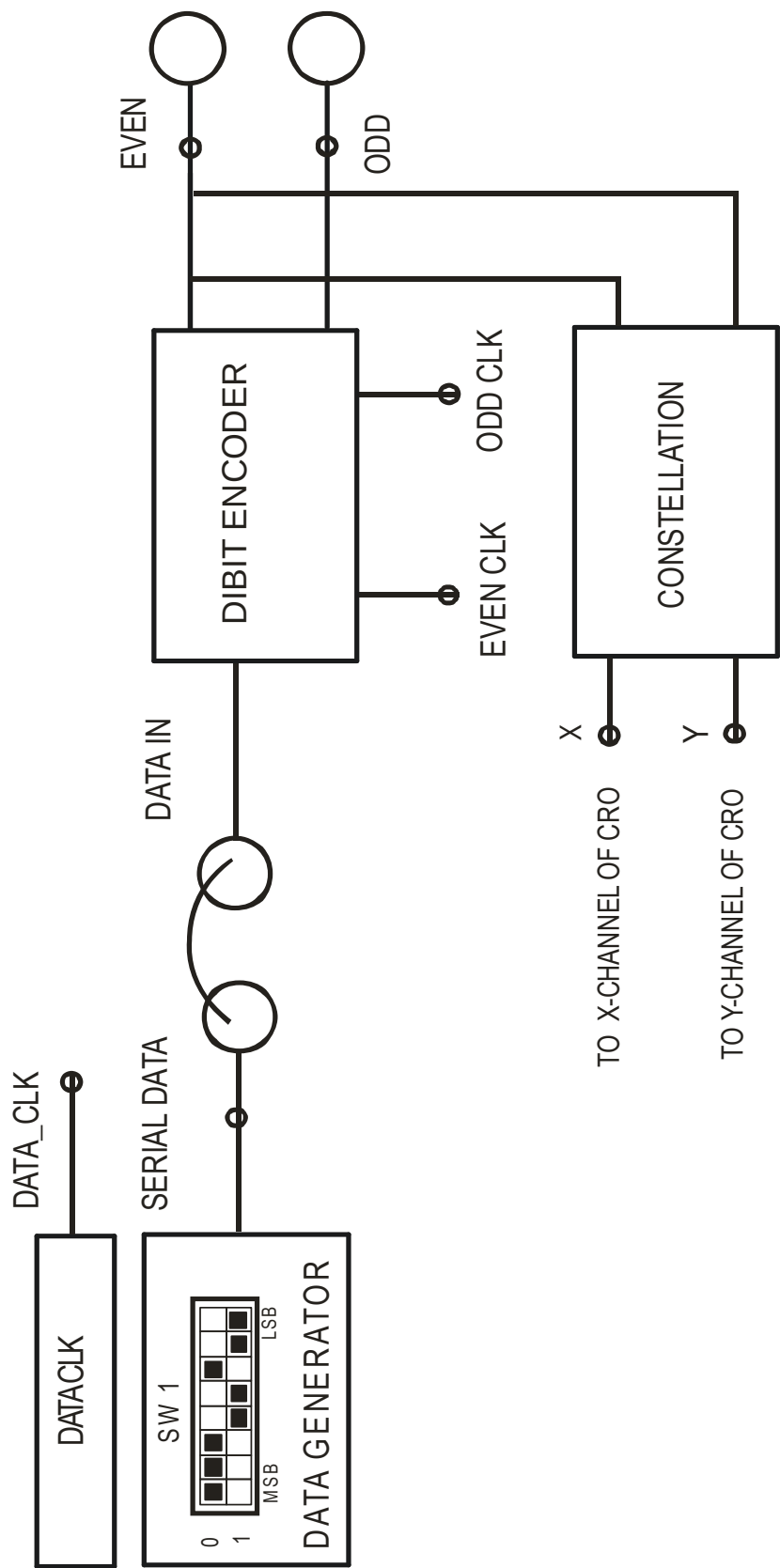


FIG 3.1 BLOCK DIAGRAM FOR OBSERVATION OF CONSTITUTION DIAGRAM

## **EXPERIMENT NO: 3**

### **NAME:**

OBSERVATION OF THE CONSTELLATION DAIGRAM FOR EVEN & ODD DATA AND STUDY OF BANDWIDTH EFFICIENCY OF QPSK.

### **OBJECTIVE:**

- (A) To study the constellation diagram of QPSK data format.
- (B) Study of bandwidth efficiency in Quadrature Phase shift keying techniques.

### **THEORY:**

The constellation diagram or geometrical representation is as shown in fig.3.2 below the points in signal space corresponding to each of the four possible transmitted signal is indicated by dots. For each such signal we can recover two bits rather than one. The distance of a signal point from the origin is  $\sqrt{E_s}$  which is the square root of the signal energy associated with the symbol, that is  $E_s = P_s \cdot T_s = P_s(2T_b)$ . As we know the ability to determine a bit without error is measured by the distance in signal space between points corresponding to the different values of the bit. We note from fig. that points which differ in a single bit are separated by the distance  $d = 2\sqrt{P_s T_b} = 2\sqrt{E_b}$  where  $E_b$  is the energy contained in a bit transmitted for a time  $T_b$ . This distance for QPSK is the same as for BPSK. Hence, altogether, we have the important result that, in spite of the reduction by factor of two in the bandwidth required by QPSK in comparison with BPSK, the noise immunity of the two systems are the same.

The number of dot points appearing in the constellation diagram depends on the number of symbol generated due to EVEN & ODD bit. The position of dot points in the quadrant of the constellation diagram is also depends on the symbol generated due to the EVEN & ODD bit as shown above. The fig below shows some examples of different constellation diagram.

In fig.3.3 the data pattern used is 01011010, if we plot the expected data pattern of EVEN & ODD bit as shown in fig, then there are 4 symbol generated as 01, 11, 10 & 00, thus there are 4 dots appearing in the constellation diagram in all four quadrant.

In fig.3.4 the data pattern used is 11011010, if we plot the expected data pattern of EVEN & ODD bit as shown in fig, then there are 3 symbol generated as 11, 01 & 10. Thus there are 3 dots appearing in the constellation diagram in the respective quadrants.

In fig.3.5 the data pattern used is 11111010. if we plot the expected data pattern of I & Q bit as shown in fig, then there are 2 symbol generated as 11 & 10. Thus there are only 2 dots appearing in the constellation diagram in the respective quadrants.

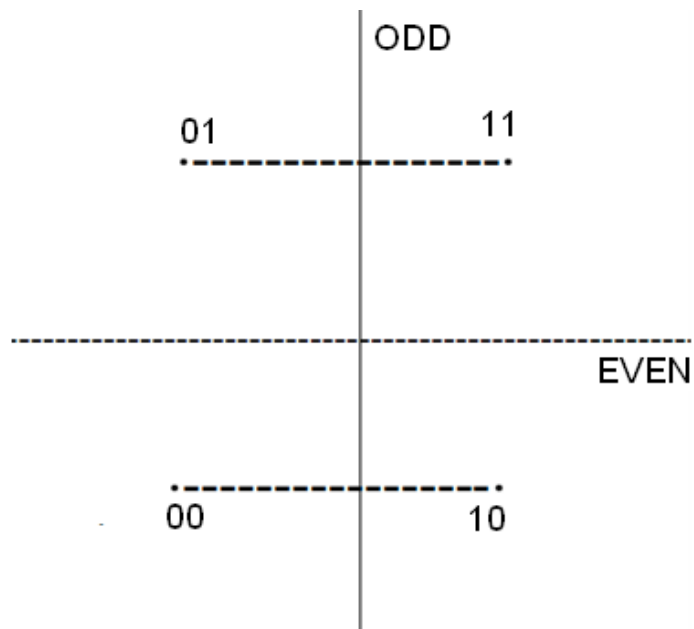


FIG.3.2 CONSTELLATION DIAGRAM

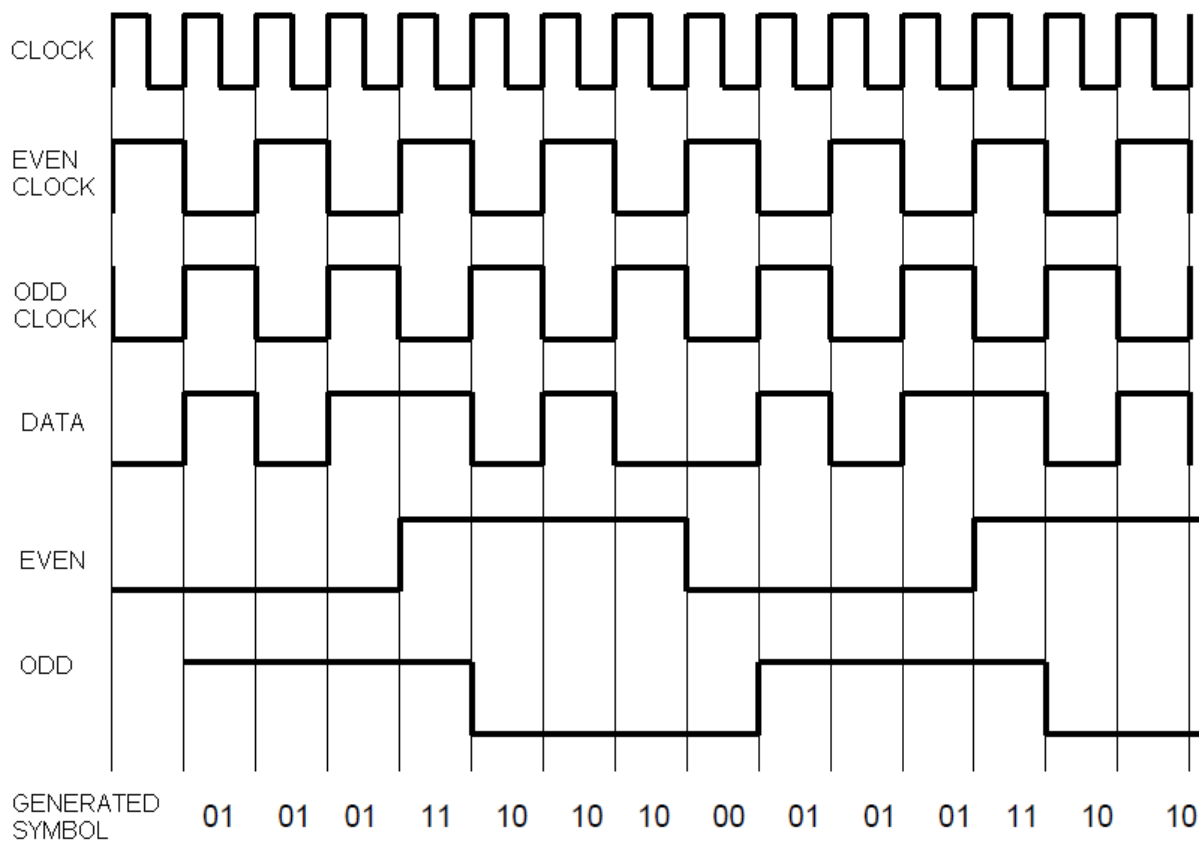


FIG. 3.3

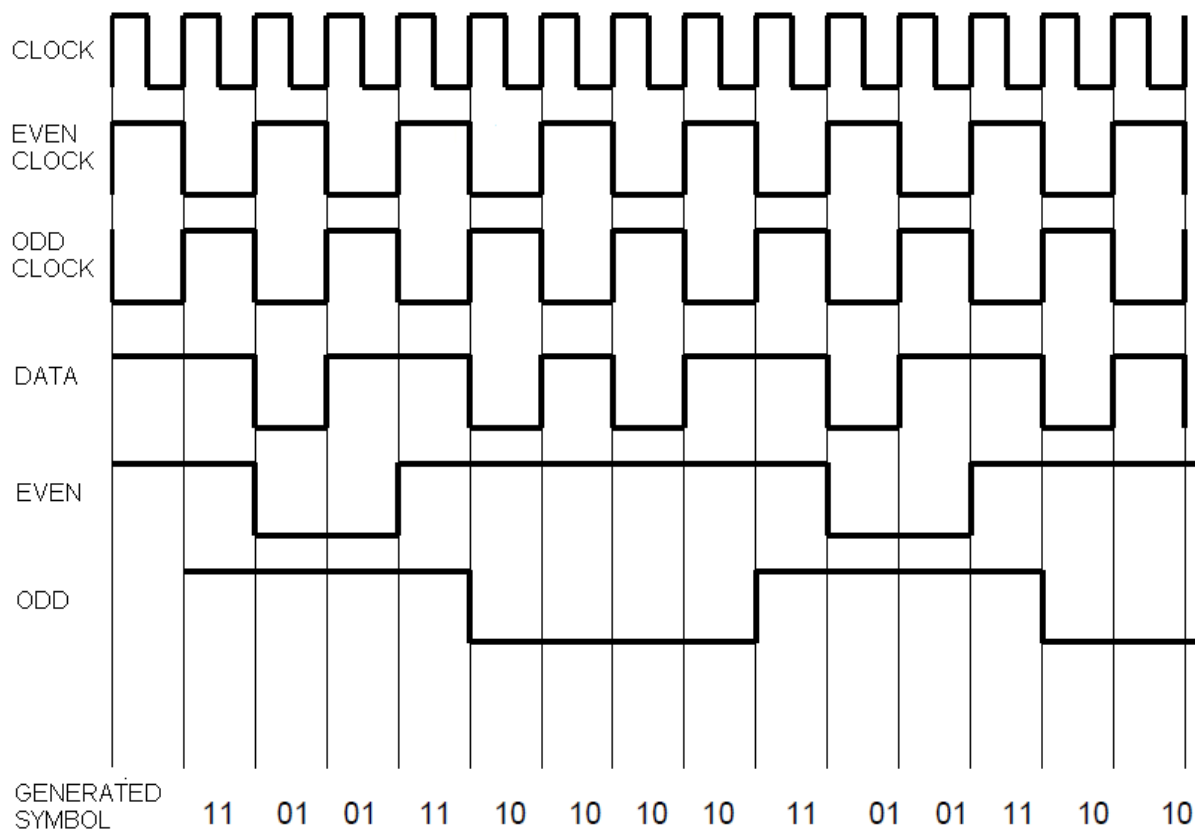


FIG. 3.4

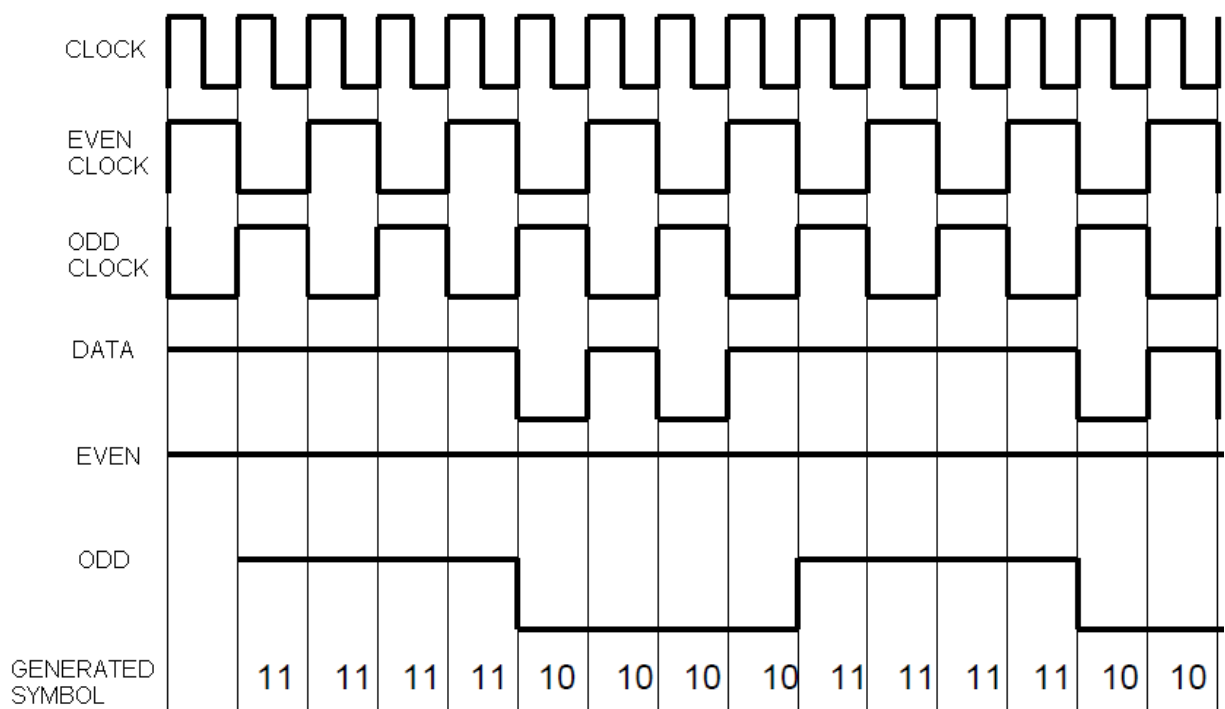


FIG.3.5

Bandwidth efficiency of QPSK is defined by the ratio of bit transmission speed in QPSK to Bandwidth of QPSK signal. It is given by

$$\text{Bandwidth Efficiency} = F_b / B_w$$

Where,

**F<sub>b</sub>** is Bit transmission speed and

**B<sub>w</sub>** is the bandwidth of signal transmitted.

The more the bandwidth efficiency more is the data transfer within the same bandwidth of signal transmission.

## **EQUIPEMENTS:**

- DCL-QPSK Kit
- Connecting Chords.
- Power supply.
- DSO.

## **PROCEDURE:** (A) Study of constellation diagram.

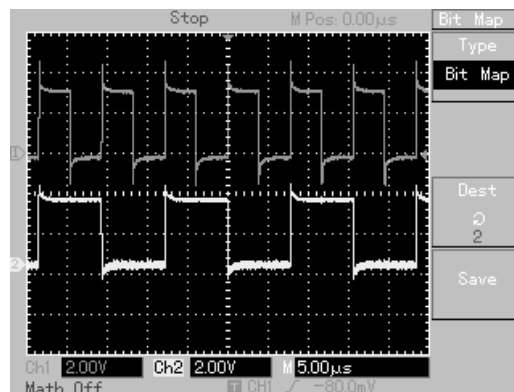
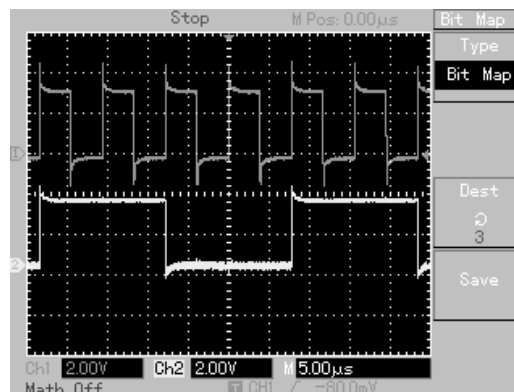
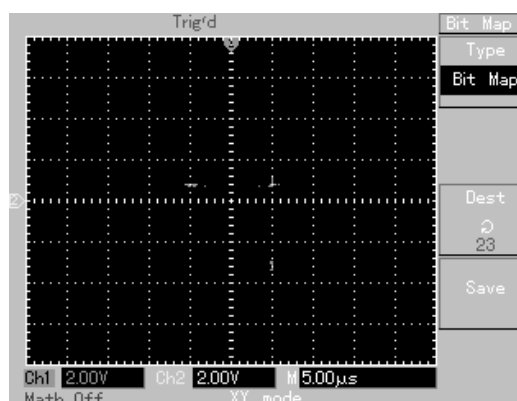
1. Refer to the block diagram (Fig.3.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **DIBIT coder**.
5. Connect the coded data **NRZ-L DATA** to the **DATA IN** of **DIBIT CONVERSION**.
6. Observe the coded signal **EVEN** data & **ODD** data bit on oscilloscope.
7. Connect **EVEN** data to X-channel of CRO & **ODD** data bit to Y-channel of CRO.
8. Put the CRO in X-Y mode. Observe the constellation diagram on oscilloscope as shown in fig.3.2 and refer theory to observe different types of constellation diagram.

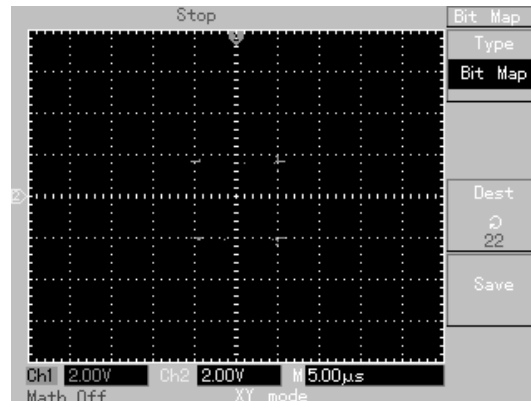
## **OBSERVATION:**

ON KIT **DCL-QPSK/DQPSK**

Observe the following waveforms on CRO and plot it on the paper.

- |                                                                   |              |
|-------------------------------------------------------------------|--------------|
| 1. Coded data <b>EVEN</b> bit with respect to <b>EVEN CLOCK</b> . | (Fig.3.6(a)) |
| 2. Coded data <b>ODD</b> bit with respect to <b>ODD CLOCK</b> .   | (Fig.3.6(b)) |
| 3. Constellation diagram on CRO.                                  | (Fig.3.6(c)) |

**CAPTURED WAVEFORM:****CH 1: EVEN CLK (128 KHz) & CH 2: EVEN DATA****Fig.3.6(a)****CH 1: ODD CLK (128 KHz) & CH 2: ODD DATA****Fig 3.6(b)****SERIAL DATA: 11011010 CH 1: EVEN & CH 2: ODD****Fig. 3.6(c)**

**SERIAL DATA: 00011011 CH 1: EVEN & CH 2: ODD****Fig. 3.6(d)****(B) Bandwidth efficiency of QPSK technique.**

1. Measure the transmission clock at **DATA\_CLOCK**post i.e. **256KHz**.
2. Divide this clock by the number of bits transmitted simultaneously, i.e. 2Bits in this case (**EVEN&ODD** data).
3. Transmission clock is the bit transmission speed (256KHz) and Bandwidth of modulator is clock divided by the number of bits transmitted (128KHz).
4. From the above data, calculate bandwidth efficiency by substituting these values in the formula as follows;

$$\text{Bandwidth Efficiency} = F_b / B_w$$

$$\text{i.e.} = 256/128$$

$$= 2$$

**CONCLUSION:**

It is observed from the constellation diagram that, for SERIAL DATA “00011011” the symbol generated due to EVEN data&ODD data have four possible values and lies in four quadrants. Also from the constellation diagram we can determine the distance between the two symbols and can conclude that after each time  $T_b$  either EVEN or ODD data can change, but both cannot change at the same time. After each time  $T_b$ , the transmitted signal, if it changes, changes phase by 90 deg. rather than by 180° as in BPSK.

Bandwidth Efficiency generally depends on the number of bits transmitted simultaneously.

# **EXPERIMENT NO. 4**

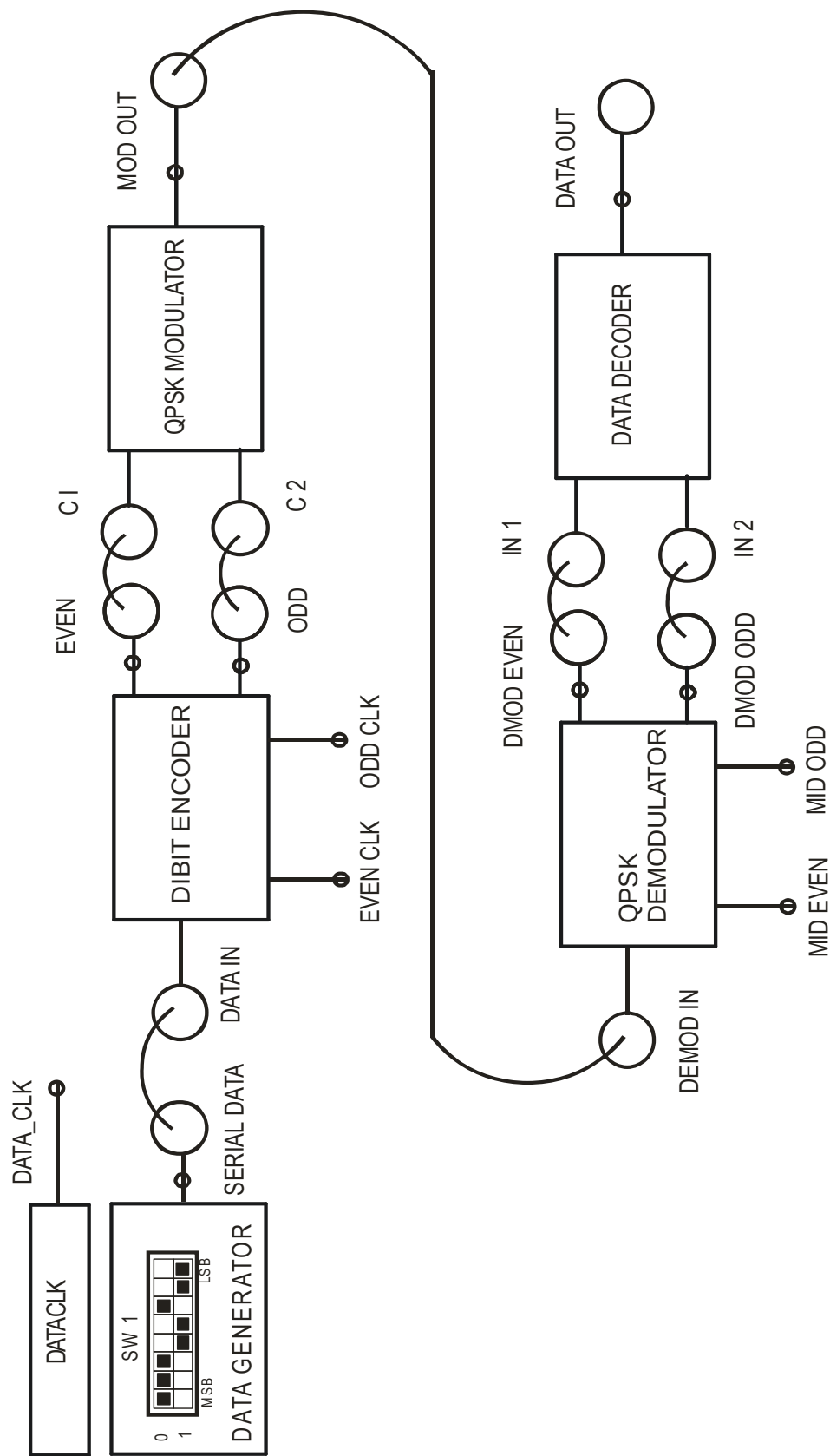


FIG 4.1 BLOCK DIAGRAM FOR  
QUADRATURE PHASE SHIFT KEYING MODULATION TECHNIQUE

## **EXPERIMENT No. : 4**

### **NAME:**

### QUADRATURE PHASE SHIFT KEYING MODULATION TECHNIQUES

### **OBJECTIVE:**

Study of Carrier Modulation Techniques by Quadrature Phase Shift Keying method

### **THEORY:**

In this modulation, called Quadrature PSK (QPSK) or 4 PSK the sine carrier takes 4 phase values, separated of 90 deg. and determined by the combinations of bit pair (Dibit) of the binary data signal. The data are coded into Dibit by a circuit generating:

- A data signal **EVEN** (in phase) consisting in voltage levels corresponding to the value of the first bit of the considered pair, for duration equal to 2 bit intervals.
- A data signal **ODD** (in quadrature) consisting in voltage levels corresponding to the value of the second bit of the pair, for duration equal to 2 bit intervals.

The block diagram of the modulator used on the module is shown in the fig.4.1 four 1MHz sine carriers, shifted between them of 90 deg, are applied to modulator. The data (signal EVEN&ODD) reach the modulator from the Dibit Encoder. The instantaneous value of EVEN and ODD data bit generates a symbol. Since EVEN and ODD can take either 0 or 1 value, maximum 4 possible symbols can be generated (00, 01, 10, and 11). According to the symbol generated one of the four-sine carrier will be selected. The relation between the symbol generated and sine carrier is shown in table.

| DIBIT |     | PHASE<br>SHIFT |
|-------|-----|----------------|
| EVEN  | ODD |                |
| 0     | 0   | 180 deg        |
| 0     | 1   | 90 deg         |
| 1     | 0   | 270 deg        |
| 1     | 1   | 0 deg          |

A receiver for the QPSK signal is shown in fig. synchronous detection is required and hence it is necessary to locally regenerate the carriers. The scheme for carrier generation is similar to that employed in BPSK. In that earlier case we squared the incoming signal to remove the phase difference, and recovered the data by filtering.

The incoming signal is applied to the multipliers to remove the phase shift. The output of the multipliers can be seen at MID EVEN and MID ODD posts. The output of the multipliers is then given to filters, where we get the recovered even and odd data at the

DEMOMOD EVEN and DEMOMOD ODD posts. These recovered EVEN&ODD bits having exactly same phase & frequency compared to transmitter EVEN&ODD bit. These EVEN&ODD bits then applied to DATA DECODER logic to recover the original NRZ-L data pattern.

## **EQUIPMENTS:**

- DCL-QPSK Kit
- Connecting Chords
- Power supply
- DSO

## **PROCEDURE:**

1. Refer to the block diagram (Fig.4.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kits **DCL-QPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **DIBIT ENCODER**.
5. Connect the dibit data **EVEN & ODD** bit to control input **C1** and **C2** of **QPSK MODULATOR** respectively
6. Connect QPSK modulated signal **MOD OUT** to the **DEMOMOD IN** of the **QPSK DEMODULATOR**.
7. Connect **DEMOMOD EVEN&DEMOMOD ODD** outputs of **QPSK DEMODULATOR** to **IN1 & IN 2** posts of **Data Decoder** respectively.
8. Observe various waveforms as mentioned below (Fig. 4.2).

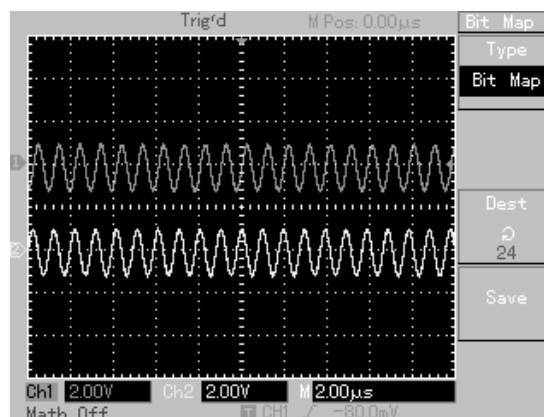
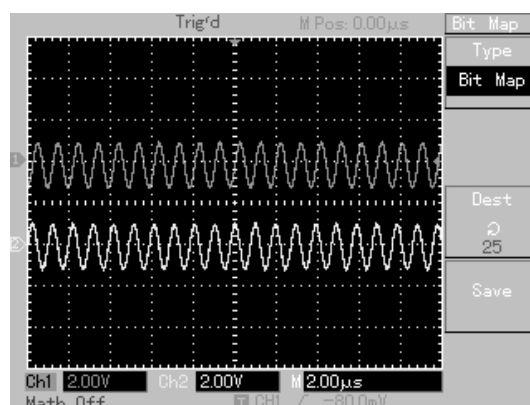
## **OBSERVATION:**

Observe the following waveforms on oscilloscope and plot it on the paper.

ON KIT **DCL-QPSK/DQPSK**

1. Input **NRZ-L** Data at **SERIAL DATA**. (Fig.4.2(a))
2. **DATA CLOCK** (Fig.4.2(a))
3. Carrier frequency **SIN 0, SIN 90, SIN 180** and **SIN 270**. (Fig.4.2(b))
4. **EVEN CLK** with respect to **ODD CLOCK**. (Fig.4.2(c))
5. Coded data **EVEN** with respect to **EVEN CLK** (Fig.4.2(d))
6. Coded data **ODD** with respect to **ODD CLK** (Fig.4.2(e))
7. QPSK modulated signal at **MOD OUT**. (Fig.4.2(f))
8. **EVEN** multiplier at **MID EVEN** with respect to **DEMOMOD EVEN** (Fig.4.2(g))
9. **ODD** multiplier at **MID ODD** with respect to **DEMOMOD ODD**. (Fig.4.2(h))
10. Recovered **DEMOMOD EVEN** with respect to Coded data **EVEN**. (Fig.4.2(i))
11. Recovered **DEMOMOD ODD** with respect to Coded data **ODD**. (Fig.4.2(j))
12. **DATA OUT** with respect to **SERIAL DATA**. (Fig.4.2(k))

## **CAPTURED WAVEFORM:**

**CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)****Fig.4.2(a)****CH 1: SIN 0 (1 MHz) & CH 2: SIN 90 (1 MHz)****Fig.4.2(b)****CH 1: SIN 180 (1 MHz) & CH 2: SIN 270 (1 MHz)****Fig.4.2(b)****CH 1: EVEN CLK (128 KHz) & CH 2: ODD CLK (128 KHz)**

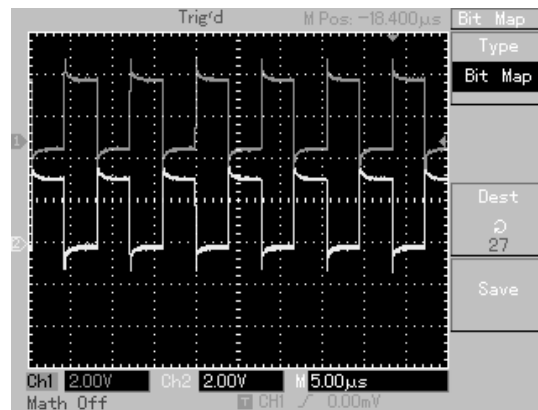


Fig.4.2(c)

CH 1: EVEN CLK (128 KHz) & CH 2: EVEN DATA



Fig.4.2(d)

CH 1: ODD CLK (128 KHz) & CH 2: ODD DATA

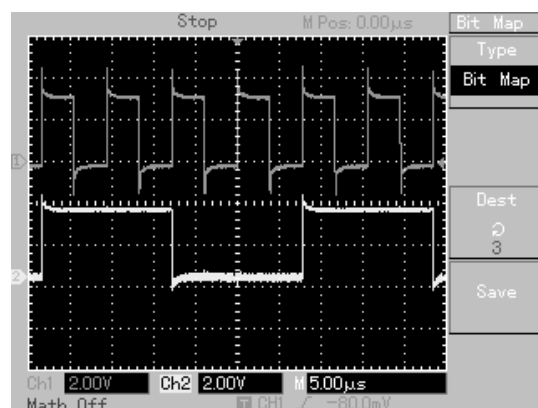
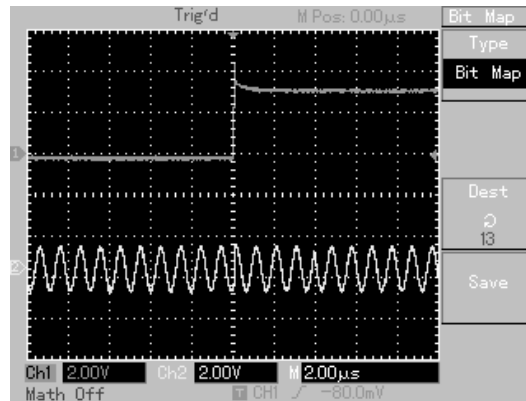
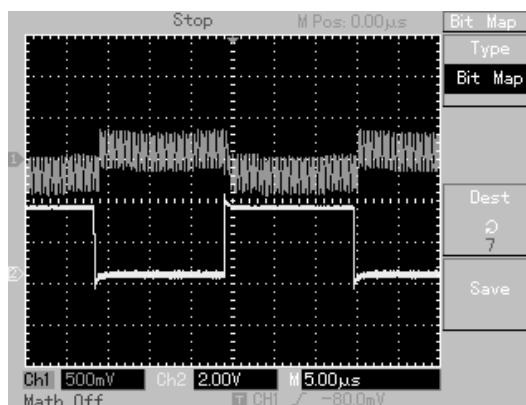
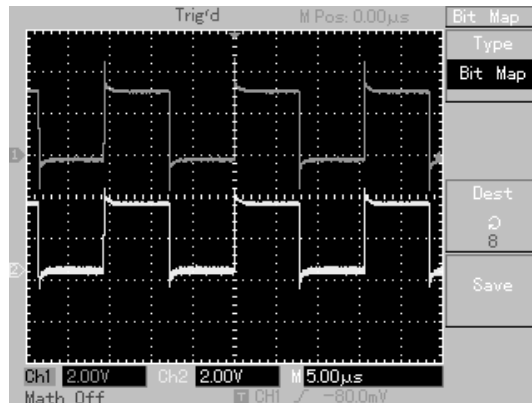
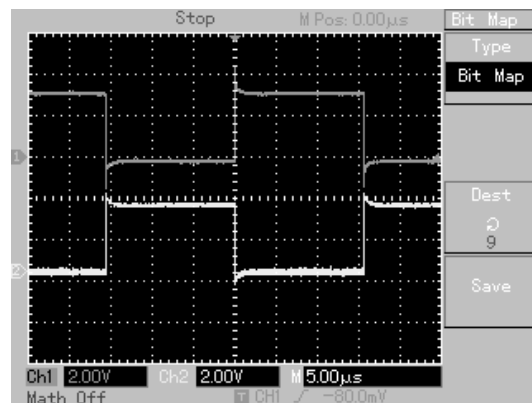
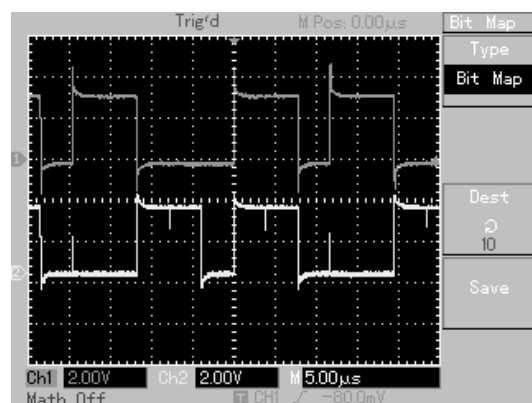


Fig.4.2(e)

#### NOTE:

For **MOD OUT**, select a symbol using **EVEN** and **ODD** and then observe the **MOD OUT** waveform below a symbol is selected using **EVEN** and **ODD** and then **MOD OUT** is observed with respect to **EVEN**.

**CH 1: EVEN & CH 2: MOD OUT****Fig.4.2(f)****CH 1: MID EVEN & CH 2: DEMOD EVEN****Fig.4.2(g)****CH 1: MID ODD & CH 2: DEMOD ODD****Fig.4.2(h)**

**CH 1: EVEN& CH 2: DEMOD EVEN****Fig.4.2(i)****CH 1: ODD& CH 2: DEMOD ODD****Fig.4.2(j)****CH 1: SERIAL DATA & CH 2: DATA OUT****Fig.4.2(k)**

## **CONCLUSION:**

In BPSK we deal individually with each bit of duration  $T_b$ . In QPSK we lump two bits together to form a SYMBOL. The symbol can have any one of four possible values corresponding to two-bit sequence 00, 01, 10, and 11. We therefore rearrange to make available for transmission four distinct signals. At the receive each signal represents one symbol and, correspondingly, two bits. When bits are transmitted, as in BPSK, the signal changes occur at the bit rate. When symbols are transmitted the changes occur at the symbol rate which is one-half the bit rate. Thus the symbol time is  $T_s = 2T_b$ .



# **EXPERIMENT NO.5**

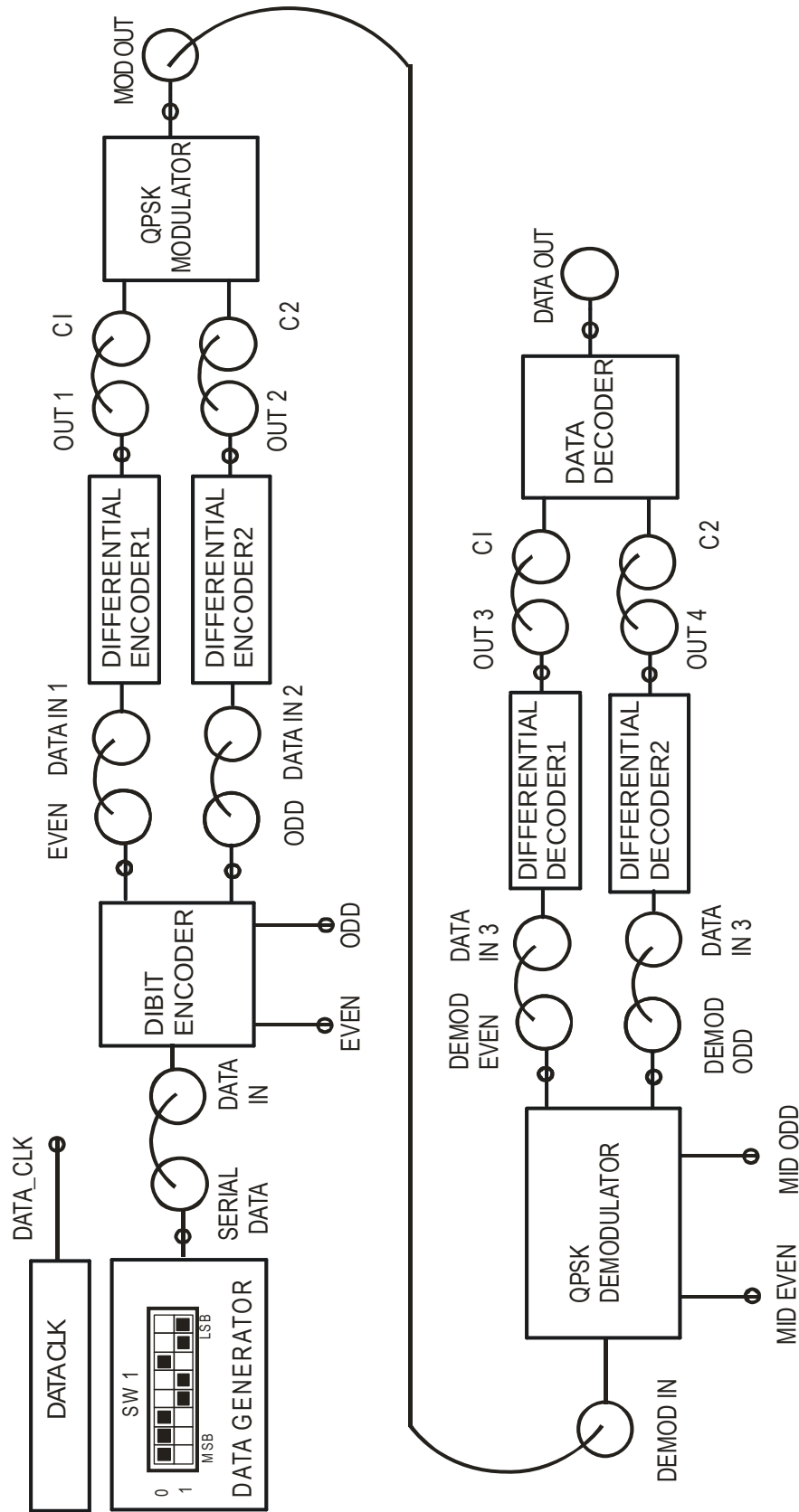


FIG 5.1 BLOCK DIAGRAM FOR  
STUDY OF DIFFERENTIAL QUADRATURE PHASE SHIFT KEYING  
TECHNIQUE

## **EXPERIMENT: 5**

### **NAME:**

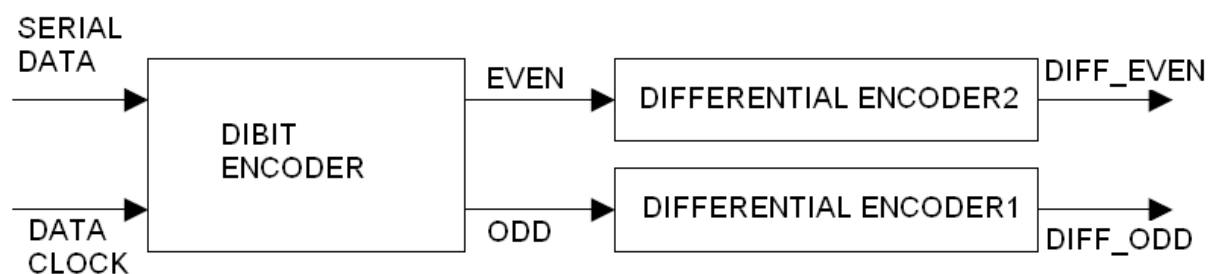
### **DIFFERENTIAL QUADRATURE PHASE MODULATION TECHNIQUE**

### **OBJECTIVE:**

Study of carrier modulation technique by Differential Quadrature Phase Shift Keying method

### **THEORY:**

In DQPSK or 4-phase PSK differential modulation, the phase of the sine carrier is changed as function of the pair of data bit (dibit). The modulator is same used for QPSK or 4-PSK absolute. Before entering the modulator the data are coded so to generate two "differential" signals namely DIFF\_EVEN and DIFF\_ODD. These consider the variation of the bit pairs, in respect to the pair just before. The block diagram of DQPSK is shown below:

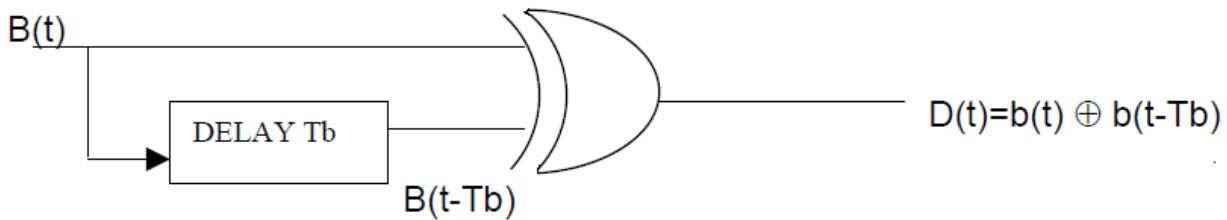


**FIG. 5.2 DIFFERENTIAL ENCODING OF DIBIT DATA**

The differential coding of data to be transmitted makes the "bit pair" to be transformed into phase variation of the carrier. In this way the receiver recognizes the pair each time it detects a phase shift of the modulated carrier, independently from its absolute phase. The ambiguity of the QPSK modulation, which can take to the demodulated data inversion, is solved in this way.

The procedure for generating the DQPSK signal is same as QPSK, but in DQPSK the EVEN & ODD bits are differentially encoded using separate Differential Encoder as shown in fig.5.2. The concept of differential encoding is same as explained in experiment no.2. After differential encoding the encoded data is fed to control inputs of QPSK MODULATOR. The four phases shifted carrier is also fed to the QPSK MODULATOR. The output of the QPSK MODULATOR is DQPSK modulated signal.

The recovery of DQPSK signal is same up to the two EVEN & ODD bit recoveries. In DQPSK we first recover the differentially encoded EVEN & ODD bits. These recovered bits need to be decoded to bring it into original EVEN & ODD bit format. Thus, two separate Differential Decoders are used at the DQPSK demodulator. The block diagram of Differential Decoder is shown in fig 5.3.



**FIG. 5.3 DIFFERENTIAL DECODER**

The logic of differential decoder is, the recovered differentially encoded EVEN & ODD bits are applied directly to one input of an exclusive-OR logic gate and to the other input is applied  $b(t-T_b)$  (see fig 5.3) i.e one bit delayed. The gate output will be at one or the other of its level depending on whether  $b(t)=b(t-T_b)$  or  $b(t) \neq b(t-T_b)$ . In the first case  $b(t)$  did not change level and therefore the transmitted bit is  $d(t)=0$ . In second case  $d(t)=1$ . At the output of differential decoder we get the original I & Q bits. These I & Q bits then applied to the data decoder along with decoder clock to recover the original NRZ-L data.

### **EQUIPMENTS:**

- DCL-QPSK Kit
- Connecting Chords.
- Power supply.
- DSO.

### **PROCEDURE:**

1. Refer to the block diagram (Fig.5.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kits **DCL-QPSK** and switch it on.
3. Select Data pattern of simulated data using switch **SW1**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **DIBIT ENCODER**.
5. Connect the dibit data **EVEN & ODD** bit to **DATA IN 1** of **DIFFERENTIAL ENCODER 1** & **DIFFERENTIAL ENCODER 2** respectively.
6. Connect **OUT 1 & OUT 2** of **DIFFERENTIAL ENCODER 1 & DIFFERENTIAL ENCODER 2** respectively to control input **C1** and **C2** of **QPSK MODULATOR**.
7. Connect DQPSK modulated signal **MOD OUT** to the **DEMOD IN** of the **QPSK DEMODULATOR**.
8. Connect **DEMOD EVEN & DEMOD ODD** outputs of **QPSK DEMODULATOR** to **DATA IN 3 & DATA IN 4** of **DIFFERENTIAL DECODER 1 & DIFFERENTIAL DECODER 2** respectively.
9. Connect **DIFFERENTIAL DECODER 1 & DIFFERENTIAL DECODER 2** outputs **OUT 3 & OUT 4** to **IN 1 & IN 2** of **DATA DECODER** respectively.

### **OBSERVATION:**

Observe the following waveforms on oscilloscope and plot it on the paper.

1. Input **NRZ-L** Data at **SERIAL DATA** (Fig.5.4 (a))
2. **DATA CLOCK** (Fig.5.4 (a))
3. Carrier frequency **SIN 0**, **SIN 90**, **SIN 180** and **SIN 270**. (Fig.5.4 (b))
4. **EVEN CLK** with respect to **ODD CLOCK**. (Fig.5.4 (c))
5. Coded data **EVEN** with respect to **EVEN CLK** (Fig.5.4 (d))
6. Coded data **ODD** with respect to **ODD CLK** (Fig.5.4 (e))
7. Differentially encoded **OUT1** with respect to coded data **EVEN** (Fig.5.4 (f))
8. Differentially encoded **OUT2** with respect to coded data **ODD** (Fig.5.4 (g))
9. DQPSK modulated signal at **MOD OUT**. (Fig.5.4 (h))
10. **EVEN** multiplier at **MID EVEN** with respect to **DEMOD EVEN**. (Fig.5.4 (i))
11. **ODD** multiplier at **MID ODD** with respect to **DEMOD ODD**. (Fig.5.4 (j))
12. Recovered **DEMOD EVEN** with respect to Coded data **OUT 1**. (Fig.5.4 (k))
13. Recovered **DEMOD ODD** with respect to Coded data **OUT 2**. (Fig.5.4 (l))
14. Recovered **OUT 3** with respect to Coded data **EVEN**. (Fig.5.4 (m))
15. Recovered **OUT 4** with respect to Coded data **ODD**. (Fig.5.4 (n))
12. **DATA OUT** with respect to **SERIAL DATA**. (Fig.5.4 (o))

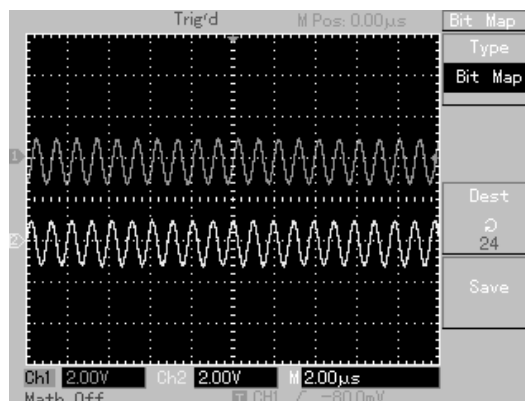
### CAPTURED WAVEFORM:

**CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)**

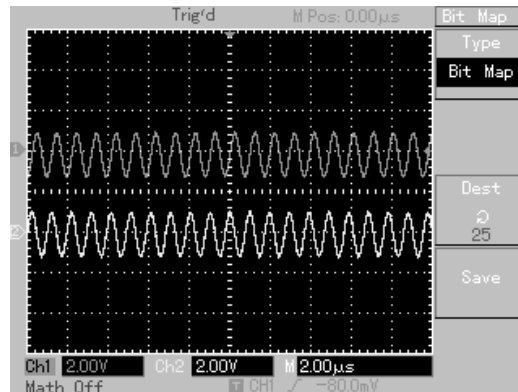
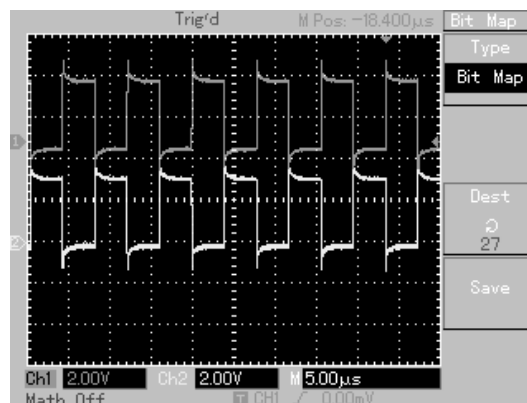


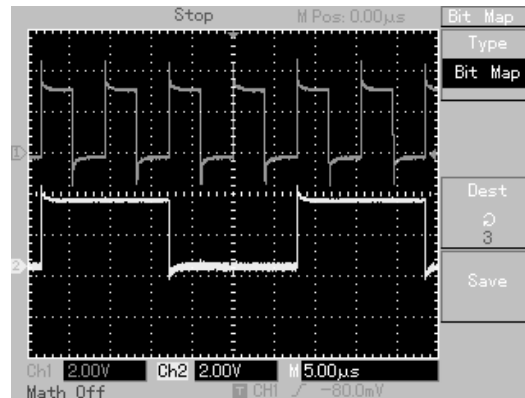
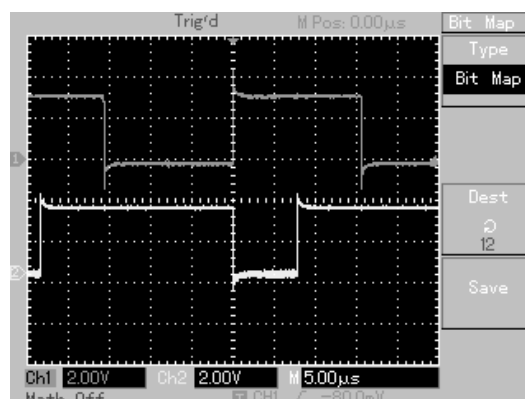
**Fig.5.4(a)**

**CH 1: SIN 0 (1 MHz) & CH 2: SIN 90 (1 MHz)**



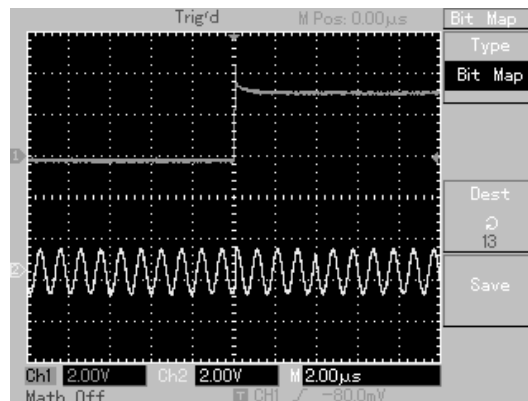
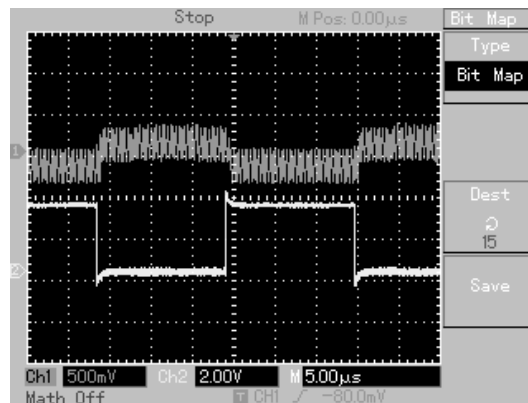
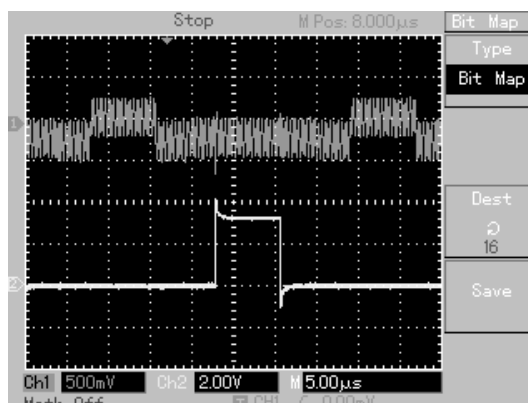
**Fig.5.4(b)**

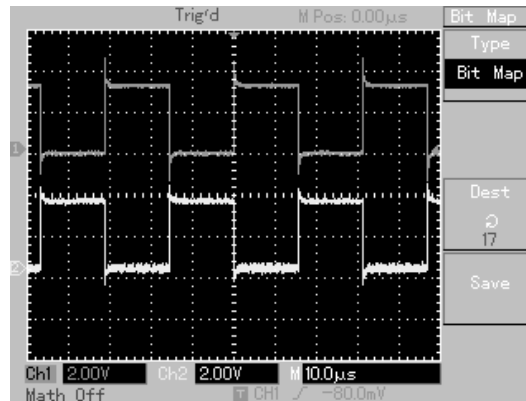
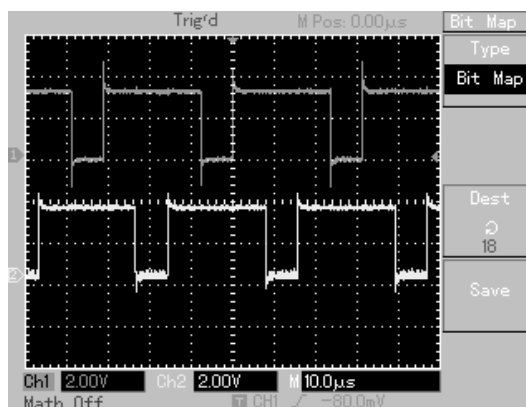
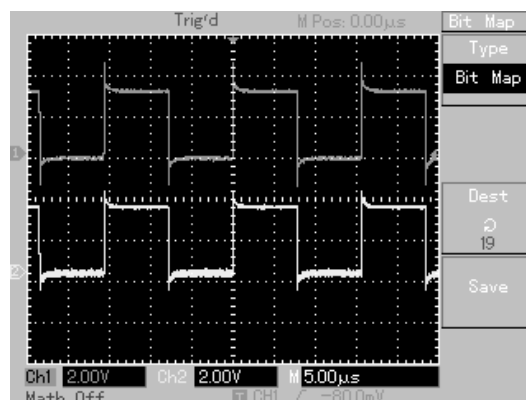
**CH 1: SIN 180 (1 MHz) & CH 2: SIN 270 (1 MHz)****Fig.5.4(b)****CH 1: EVEN CLK (128 KHz) & CH 2: ODD CLK (128 KHz)****Fig.5.4(c)****CH 1: EVEN CLK (128 KHz) & CH 2: EVEN DATA****Fig.5.4(d)**

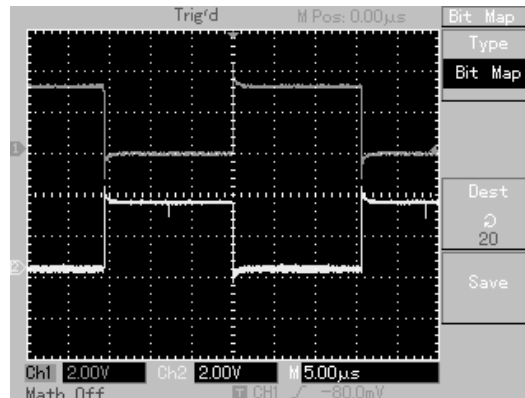
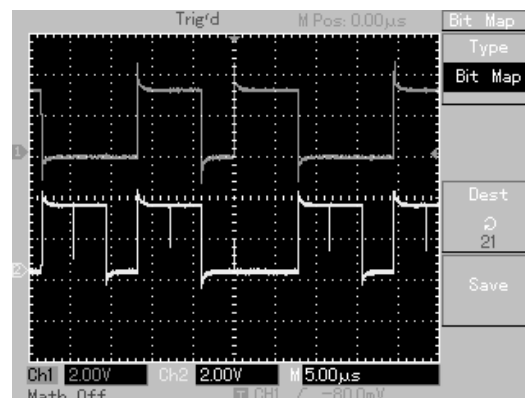
**CH 1: ODD CLK (128 KHz) & CH 2: ODD DATA****Fig.5.4(e)****CH 1: EVEN DATA & CH 2: OUT 1****Fig.5.4(f)****CH 1: ODD DATA & CH 2: OUT 2****Fig.5.4(g)**

**NOTE:**

For **MOD OUT**, select a symbol using **OUT 1** and **OUT 2** and then observe the **MOD OUT** for waveform below a symbol is selected using **OUT 1** and **OUT 2** and then **MOD OUT** is observed with respect to **OUT 1**.

**CH 1: OUT1 & CH 2: MOD OUT****Fig.5.4(h)****CH 1: MID EVEN & CH 2: DEMOD EVEN****Fig.5.4(i)****CH 1: MID ODD & CH 2: DEMOD ODD****Fig.5.4(j)**

**CH 1: DEMOD EVEN & CH 2: OUT 1****Fig.5.4(k)****CH 1: DEMOD ODD & CH 2: OUT 2****Fig.5.4(l)****CH 1: EVEN & CH 2: OUT 3****Fig.5.4(m)**

**CH 1: ODD & CH 2: OUT 4****Fig.5.4(n)****CH 1: SERIAL DATA & CH 2: DATA OUT****Fig.5.4(o)****CONCLUSION:**

The differential coding of EVEN&ODD bit data to be transmitted makes the “bit pair” to be transformed into phase variation of the carrier. In this way the receiver recognizes the pair each time it detects a phase shift of the modulated carrier, independently from its absolute phase. The ambiguity of the QPSK modulation, which can take to the demodulated data inversion, is solved in this way.